Seeing Through Haze, Water, Glass and Plastic

Can classical image processing restore the picture? A simple walk-through

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4 November 2026 (presentation 12 November)

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Abstract

We take pictures that were spoiled by haze, murky water, rain on glass or cloudy plastic, and try to repair them using classical image-processing tools only (no machine learning). The well-known Dark Channel Prior (DCP) is our baseline. Because real spoiled pictures with a known "clean" answer are rare, we also *make* spoiled pictures ourselves ([SYNTHETIC]). A large part of this report asks: *can the made-up pictures be trusted to predict what happens on real ones?* The short answer: partly. They agree well for glass; for haze the full ranking of methods agrees only moderately (rank correlation +0.30 with our calibrated haze and +0.57 with the original haze on 90 scenes, and only +0.09 on outdoor scenes), though the verdict about DCP is the same.

How to read the tags. Every figure and table says where its pictures came from: [SYNTHETIC] pictures we made from clean photos (the true answer is known exactly), [REAL] photographs that were really spoiled and have a clean twin, or [REAL vs SYNTHETIC] a comparison between the two.

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1 The question, in plain words

A hazy photo looks grey and flat. A photo through rain-spotted glass has blurry blobs. Underwater photos look blue-green. Can we undo this with the tools taught in a first image-processing course: stretching contrast,

equalising histograms, filtering, working in the frequency domain, and so on?

We ask four things:

1. What does each "spoiling medium" actually do to light?
2. Which classical method repairs each medium best, and does anything beat DCP?
3. Does the order of steps matter (for example, sharpen then equalise, or equalise then sharpen)?
4. Can pictures we make by computer stand in for real ones?

How the report builds up. Each section uses only what the earlier ones introduced.

2	**What a spoiling medium does to light**: one equation, then each medium and its frequency signature.
3	**The pictures we use**: made-up (synthetic) and real, and why we need both.
4	**The repair methods**: every technique explained in a line, the pipelines (flow diagrams and pictures of each step), and how we score.
5	**Results on made-up pictures**, where the true answer is known.
6	**Results on real pictures**, haze, raindrops and underwater.
7	**Can made-up pictures stand in for real ones?** Seven tests.
8	**Global or local**: should a method act on the whole picture or tile by tile?
9	**Why it works this way**, then conclusions and limits.

2 What a spoiling medium does to light

2.1 One idea covers almost everything

Light from the scene passes through the medium on its way to the camera. Two things happen:

- the medium **lets only a fraction t of the scene light through** (so the scene looks fainter), and
- the medium **adds light of its own**, a milky glow of brightness A.

Written as an equation, with J the clean scene and I the picture we see:

$$I \;=\; t\,J \;+\; A\,(1-t). \tag{1}$$

If $t = 1$ nothing is lost. If t is near 0 we see only the glow A. In fog, t gets smaller the farther away something is: $t = e^{-\beta d}$ with distance d and fog thickness β.

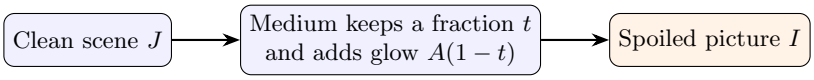


Figure 1: The one idea behind haze, water, glass film and plastic. What differs is where t and A come from.

2.2 The four media

Medium	What it does	Difference from plain haze
Haze	Fraction t falls with distance; grey-white glow.	The base case.
Water	Same, but red light dies first, so far things turn blue-green.	Each colour has its own t; the glow is dark blue-green, not bright.
Glass	A milky film, smudges, and *raindrops*. Drops act like tiny lenses that replace small patches of the scene.	Raindrops are patches, not a smooth veil.
Plastic	Dims, tints (often yellow), *blurs* slightly, and adds a veil.	The blur is new; it is the same everywhere (not distance-dependent).

The same four media in the literature's terms.

Medium	Typical cause	Visual result	Standard term
Air	smoke, dust, fog, humidity	grey or blue wash, distant detail lost	**atmospheric attenuation** (scattering and airlight)
Water	algae, sand, plankton, floating particles	green or blue tint, strong contrast loss	**turbidity** and **backscatter**
Plastic	polymer crystals, surface scratches	milky, cloudy or semi-opaque	**material haze** (ASTM D1003: the share of transmitted light scattered by more than 2.5 degrees)
Glass	smudges, dust, condensation, texture	halos around lights, washed-out contrast	**veiling glare** and **smudge flare**

In our equations: atmospheric attenuation and backscatter are both the $tJ + A(1-t)$ of Section 2; our plastic model (blur, tint and a milky veil) stands in for material haze; our glass film and smudge stand in for veiling glare.

When two media are stacked (say, murky water seen through plastic), the equation is simply applied twice: $I_2 = t_1 t_2 J + t_2 A_1 (1 - t_1) + A_2 (1 - t_2)$.

2.3 What this means for frequencies (a second view)

Splitting a picture into coarse and fine detail (the Fourier transform) gives a neat summary of what each medium does:

- **Haze:** every level of detail is multiplied by t; only the average brightness gets the glow. Haze therefore *fades detail evenly.*
- **Plastic:** fades detail like haze, *and* kills fine detail more than coarse detail (blur).
- **Water:** like haze but with a different t for red, green and blue.
- **Stacks:** the effects multiply.

We checked these predictions on our made-up pictures: they hold essentially exactly (Appendix A). The full derivations, with a literature check, are in `docs/model-equations.md`.

3 The pictures we use

We need pairs: a spoiled picture and the clean picture it came from, so we can score a repair.

	[SYNTHETIC] made by us	**[REAL]** photographed
Haze	Clean indoor photos (NYU Depth v2, 24 rooms) with their measured depth are fogged with the equation above; 3 strengths.	O-HAZE, I-HAZE, Dense-Haze, NH-Haze (a fog machine was used and the same scene photographed with and without): 185 pairs.
Water	Same, with per-colour attenuation.	(a) TUB: 1,183 pairs from a water tank in which milk was added in 10 steps, camera and objects fixed, so each turbid picture has a near-clear twin (the least turbid picture of the same view). (b) UIEB: 150 ocean and reef photos with *no* clean twin, so only reference-free scores.
Glass	Same, rain drops and film.	RainDrop dataset (58 photos of a pane with drops and the same pane clean).
Plastic	Same, blur and tint.	**None exists.** Own captures are planned.

Why use made-up pictures at all? Their true answer is exact, we can make as many as we like at any strength, and we can test one thing at a time.

What is the catch? Our equation is a model: if it is wrong, methods may look good on made-up pictures and still fail on real ones. Section 7 measures how big that catch is.

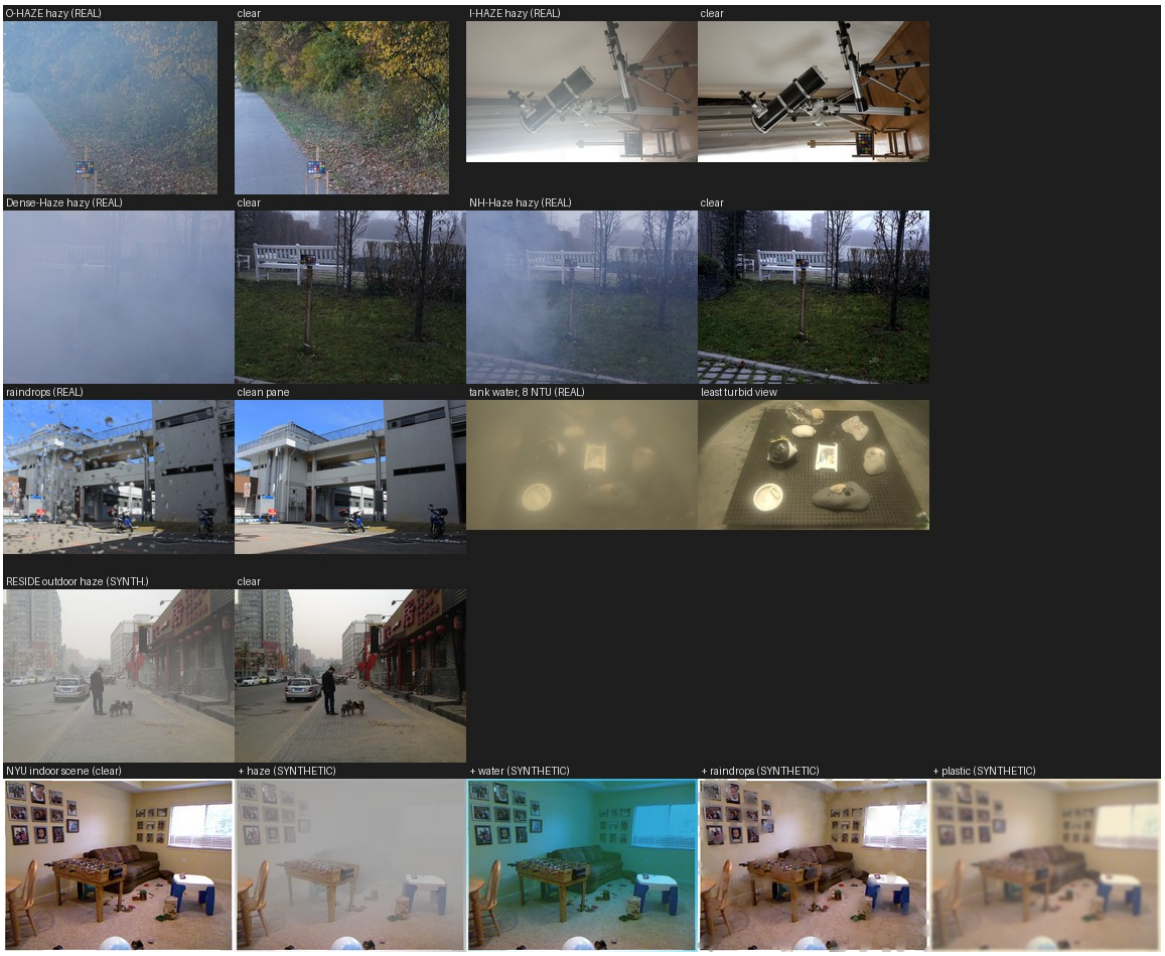


Figure 2: A sample from each source (spoiled picture, then its clear twin where there is one). Rows 1 to 2: real haze (O-HAZE, I-HAZE, Dense-Haze, NH-Haze). Row 3: real raindrops and real tank water. Row 4: outdoor haze made by the RESIDE authors. Row 5: one clear NYU indoor scene and the same scene degraded by each of our four models ([SYNTHETIC]). The UIEB ocean photos are shown in Section 6.

Medium	Cause	Data we have	Coverage	More datasets found (not downloaded)
Air	fog	O-HAZE, I-HAZE, Dense-Haze, NH-Haze (real fog-machine pairs, 185); RESIDE (outdoor, made)	covered	–
Air	smoke	NH-Haze (uneven machine fog, as a stand-in)	partly	A2I2-Haze: real smoke-machine pairs from the air and the ground, with in-place smoke measurements; behind a login and a consent notice, terms not stated
Air	dust, humidity	none	missing	none found
Water	turbidity, backscatter	TUB tank pairs (milk, near-clear reference)	covered, with a caveat (milk)	TURBID (milk, chlorophyll, blue-dye subsets with a clear reference; source not found); UFEN (talcum powder, with turbidity readings); SOTRUE (stereo, turbid; not checked)
Water	green or blue tint	UIEB ocean photos, no reference	partly	TURBID blue-dye subset (clear reference)
Water	algae, sand, plankton	none	missing	TURBID chlorophyll subset (closest to algae); UFEN talcum powder (as fine sediment)
Plastic	material haze, scratches	none (synthetic model only)	missing	none public; papers use ground-glass diffusers with a clear reference but release no data
Glass	raindrops	RainDrop pairs (58)	covered	Eigen et al. 2013 (dirt and rain on windows); "Seeing through Unclear Glass" (release unconfirmed)
Glass	smudges, dust, condensation	none (synthetic film and smudge)	missing	Flare7K++ (962 real flare photos taken through dusty and scratched lenses; non-commercial; Google Drive and Baidu)
Glass	halos around lights	none	missing	Flare7K++ (night-time flare)
Stacks	several media at once	none (synthetic only)	missing	none found

Table 1: Coverage of the causes in the table above by the data we have, and what else exists. "Covered" means real pictures with a clean reference.

source	kind	reference	available	scored in the main comparison	other analyses
NYU indoor scenes	clear scenes with depth (we degrade them)	–	150	12 scenes x 3 severities	up to 90 scenes
O-HAZE	real haze pairs	clear photo	45	in the 185	all
I-HAZE	real haze pairs	clear photo	30	in the 185	all
Dense-Haze	real haze pairs	clear photo	55	in the 185	all
NH-Haze	real haze pairs	clear photo	55	in the 185	all
all real haze		clear photo	185	185	23 to 46 per chain test
RainDrop test_a	real raindrops	clean pane	58	58	58 (same-scene twins)
UIEB	real ocean photos	none	150 (of 890)	36 (no-reference scores)	150 (figures)
TUB tank	real turbid water pairs	least-turbid view	1183	120	228 (chain tests)
RESIDE-SOTS outdoor	haze made by RESIDE	clear photo	492	–	100 (outdoor-haze test)

Table 2: Which pictures exist and which are actually scored. "Main comparison" is the first method-by-method comparison (Section 5 and 6); the last column lists the largest other uses.

4 The repair methods and how we score them

4.1 The baseline: Dark Channel Prior (DCP)

In most clean outdoor photos, a small patch contains at least one very dark pixel. In haze, that pixel is lifted by the glow. So the *darkest value in each patch* tells us how thick the haze is there. DCP uses this to guess A and t and then solves the equation backwards: $J = (I - A)/t + A$.

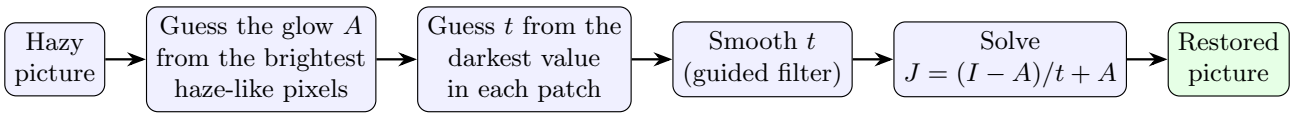


Figure 3: DCP, the baseline.

DCP assumes bright glow and dark pixels in clear scenes. That is a good match for haze and a poor one for water (dark blue-green glow) or plastic.

4.2 The tools, one by one

Every method below takes one picture and returns one picture. In the experiments they are called by the short names in the first column. The third column says what you *see* and what the method *assumes*: when the assumption is false, the method fails.

Name	What it does	Visible effect and assumption
Brightness and contrast (act on the brightness values only)		
`stretch`	Maps the darkest 1% of values to black and the brightest 1% to white, linearly in between.	Fills the full range. Assumes the picture is a dimmed version of a good one.
`hist_eq`	Histogram equalisation of the lightness: spreads the brightness values evenly over the range.	Strong contrast, can look harsh. Assumes an even spread of brightness is a good picture.
`hist_eq_rgb`, `eq_hsi`, `eq_avg_hist`	The same, but per colour channel, on the intensity only (keeps hue), or towards the average of the three channel histograms.	Per-channel can shift colours; the others keep colours steadier.
`hist_spec`	Histogram specification: reshapes the histogram into a chosen bell curve.	Natural-looking mid-tones.
`clahe`	Contrast-limited adaptive equalisation: equalises each tile of the picture separately (with a limit on how much it may boost) and blends.	Local contrast and detail. Can add noise or halos.
`gamma_auto`, `power_law`	A power curve I^γ; `gamma_auto` picks γ so the average brightness becomes 0.5.	Brightens dark pictures without clipping. Assumes the picture is just too dark.
`log`	$\log(1+I)$: stretches the dark values, squeezes the bright ones.	Lifts shadows.
Colour correction		
`gray_world`	Scales each colour so the picture's average is grey.	Removes a tint. Wrong when the scene is really one colour.
`white_patch`	Scales each colour so its brightest values become white.	Removes a tint. Fooled by one coloured highlight.
`shades_of_gray`	Between the two above: estimates the light colour with a power mean.	A compromise.
`von_kries`	White balance done on the eye's cone responses instead of red-green-blue.	Natural-looking neutral colours.
`red_comp`	Adds red back where red is weak: $R' = R + (\bar{G} - \bar{R})(1-R)G$.	Undoes the loss of red in sea water. Assumes red is the missing colour.
Haze and veil removal		
`dcp`, `dcp_guided`	Dark Channel Prior (Section 4.3); the guided version smooths the transmission to follow edges.	Removes haze. Assumes dark pixels exist and the glow is bright.
`wb_dcp`	Gray-world white balance, then DCP.	Helps when a colour cast confuses DCP.
`udcp`	Underwater DCP: the dark channel ignores red.	Water; red is already gone.
`dcp_near_global`, `dcp_local_airlight`	DCP with one almost constant transmission, or with a glow that changes across the picture.	Glass: the film is on the pane, not at a distance.
`dcp_clahe`	DCP followed by CLAHE.	Recovers contrast flattened by DCP.
`tarel`	Median filters estimate the veil; subtract it and lift the tones.	Fast fog removal without a dark-channel assumption.
`lowfreq_subtract`, `blind_veil`	Subtract the slow-changing part of the picture, or estimate t from detail strength and a local veil, then invert.	Haze removal by subtraction (Section 4.3).
`retinex`	Multi-scale Retinex: each pixel relative to its blurred surroundings at three scales.	Evens out lighting; can look unnatural.
Uneven lighting or film		
`flat_field`	Estimates the slow brightness variation with a very wide blur and divides it out.	Removes a smooth film or shading.
`homomorphic`	Takes the log, turns down slow variations and up fast ones in the frequency domain, undoes the log.	Evens lighting while keeping detail.
`top_hat`	Removes the background with a morphological opening and keeps what sticks out.	Removes broad shading; keeps small bright detail.
`local_stats`	Enhances only where the local average is dark and the local spread is small.	Lifts dull regions only.
Sharpening and deblurring (frequency or spatial)		
`unsharp`	Adds back (picture − its blurred copy).	Crisper edges; amplifies noise.

Name	What it does	Visible effect and assumption
`freq_laplacian`	Boosts high frequencies in the Fourier domain with gain growing with frequency.	Sharpening in the frequency domain.
`wiener`, `cls`	Undo a known Gaussian blur in the Fourier domain, with a noise term so noise is not blown up.	Sharper, if the blur size is right.
`richardson_lucy`	Iterative deblur: compare the blurred estimate with the picture and correct, 12 rounds.	Sharper, keeps values positive.
`median`, `adaptive_nr`	Median filter; local noise reduction that smooths where the variation looks like noise.	Removes specks and noise.
Fusion		
`fusion`	Blends three enhanced copies (white-balanced, plus CLAHE, plus brightened) level by level through a Laplacian pyramid.	Balanced result, but can over-correct colour.
`fusion_dcp`	Blends the original and three versions of the DCP result (Section 4.3).	Keeps DCP's haze removal and recovers detail.
`none`	Does nothing.	The score every method must beat.

A *sequence* chains methods: `A > B > C` means "run A first, then B on A's output, then C on B's output". Order matters (Section 4.3 shows why).

4.3 The pipelines, medium by medium

A *pipeline* is the ordered list of steps applied to one picture. Below, each box is one step; the pictures that follow show what each step does to a real or synthetic picture. The step names match the code in `code/restore/`.

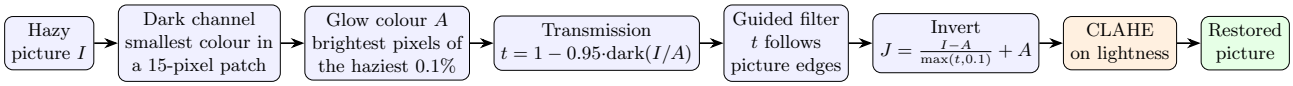


Figure 4: Guided DCP followed by CLAHE ("DCP then CLAHE"), the best single chain on real haze. The orange box is the added post-processing step.

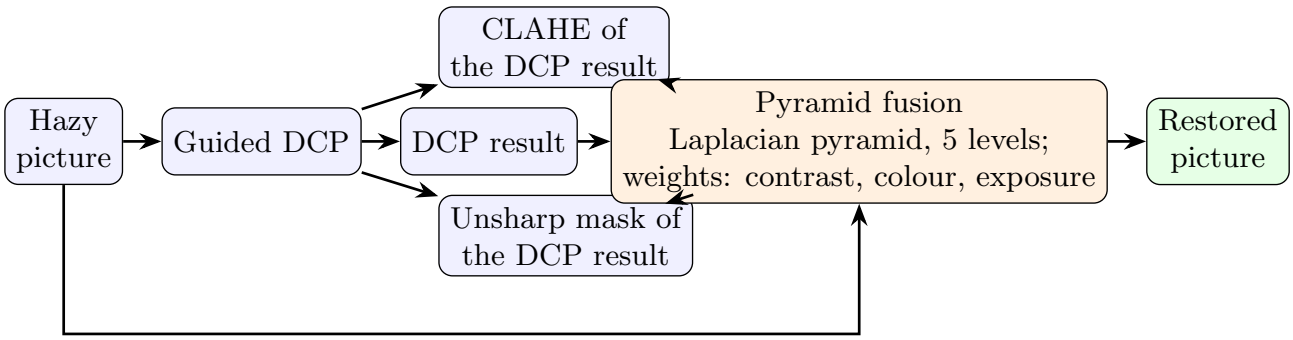


Figure 5: `fusion_dcp`: the original and three versions of the DCP result are blended, level by level, keeping at each place the version with the best contrast, colour and exposure.

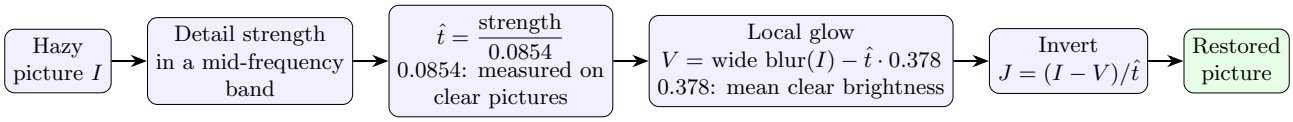


Figure 6: Haze removal by subtraction (`blind_veil`, local form). No dark-channel assumption: it uses the frequency result that haze scales detail by t.



Figure 7: [REAL] The haze pipelines on one real photo. Top: hazy photo; step 1 dark channel (bright = hazy); step 2 transmission t (dark = thick haze); step 3 the DCP result (dark, blue cast). Bottom: DCP then CLAHE; fusion; local-glow subtraction; the clear photo.

Haze (the baseline and what we add to it).

The longer chains, step by step. A chain is only worth keeping if each step pays for itself. The figure shows the best haze chain after every step, with the PSNR under each picture. On the [SYNTHETIC] haze the whole chain gains 2.7 dB. On the [REAL] photo (second row) equalisation alone makes it *worse* (11.0 to 9.2 dB), CLAHE and DCP then recover it to 12.4: the later steps repair what the first one broke. The third row is the best real-haze chain, DCP then CLAHE.

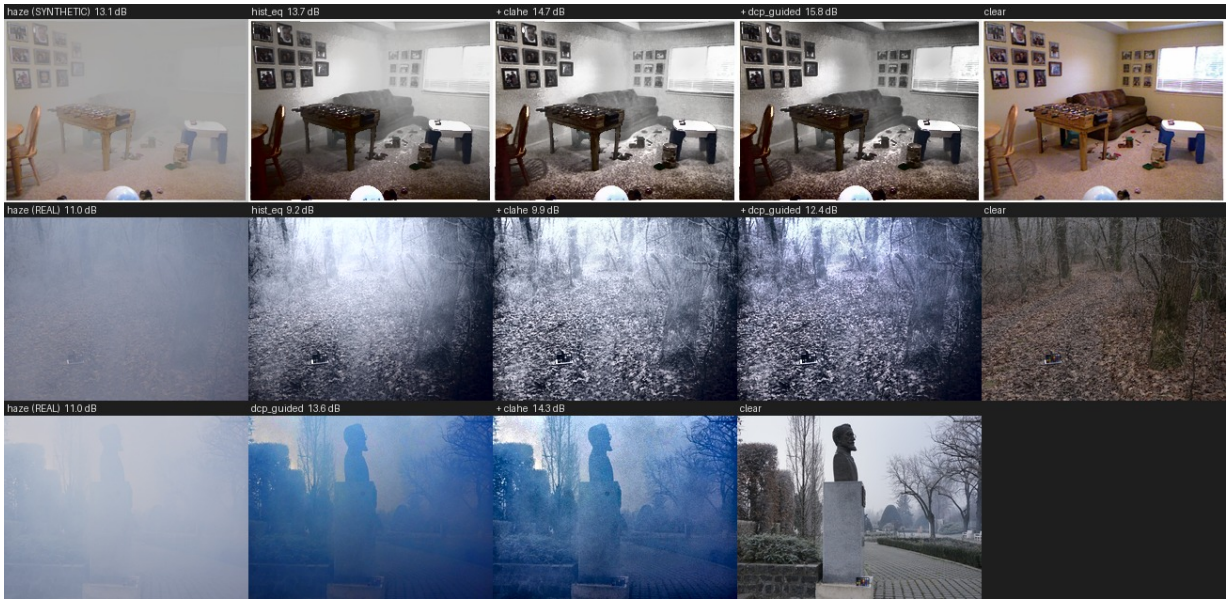


Figure 8: The haze chain `hist_eq > clahe > dcp_guided` on [SYNTHETIC] haze (row 1) and on a [REAL] photo (row 2); row 3: [REAL] photo with `dcp_guided > clahe`. The PSNR against the clear picture is in each caption.

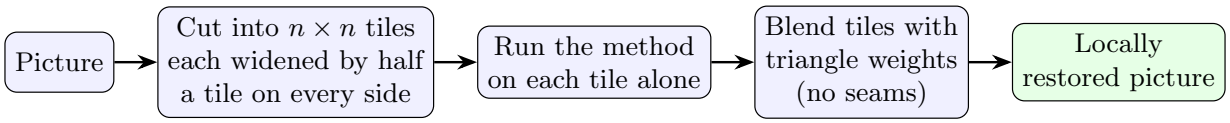


Figure 9: Tile-wise application (Section 8). The variant with regions groups pixels by haze thickness (dark channel) or sharpness instead of cutting a grid.

Local application (a wrapper that works with any method).

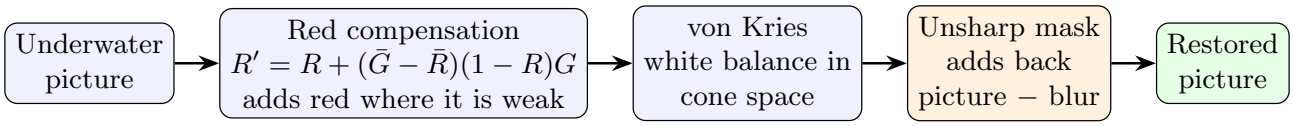


Figure 10: Best chain on synthetic water (`red_comp`, `von_kries`; the unsharp step is the searched third step). UDCP differs from DCP only in using green and blue for the dark channel, because red is already gone.

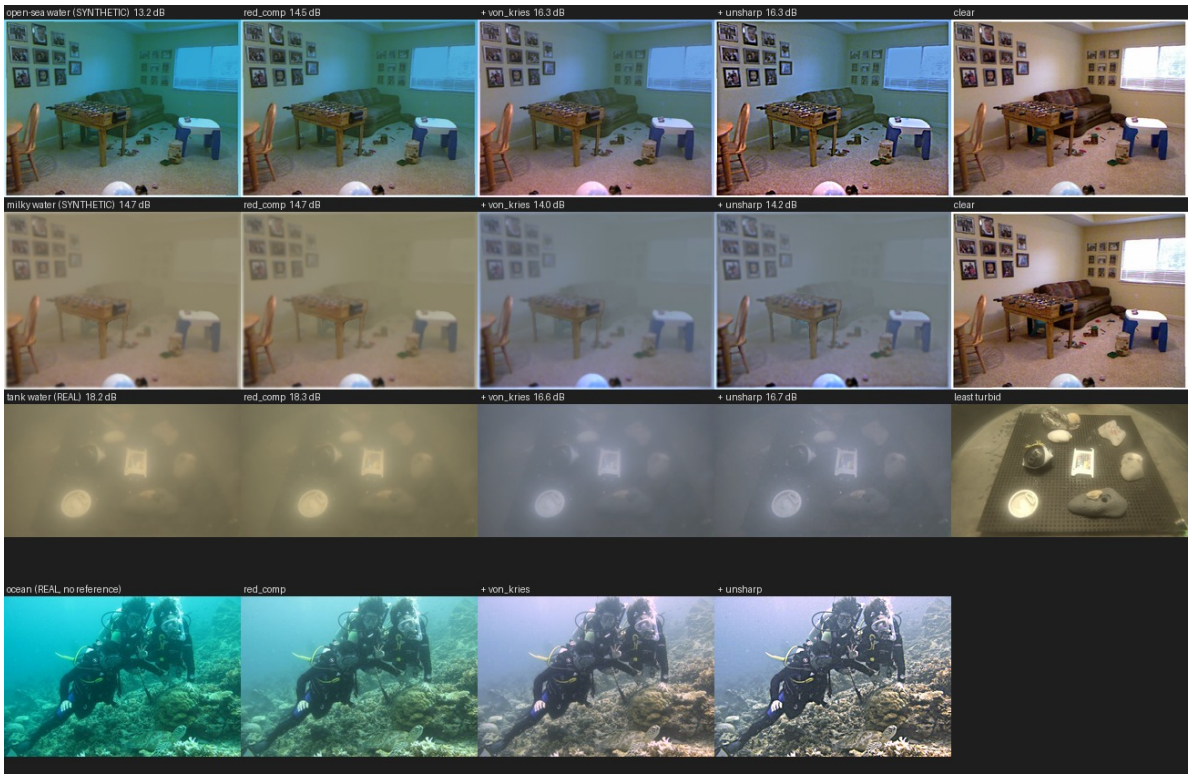


Figure 11: The chain `red_comp > von_kries > unsharp` on four kinds of water. Row 1: [SYNTHETIC] open-sea water (+3.1 dB). Row 2: [SYNTHETIC] milky water (worse: 14.7 to 14.2 dB). Row 3: [REAL] tank water (worse: 18.2 to 16.7 dB). Row 4: [REAL] ocean photo (no reference, so no score). Where the water really has a colour cast (rows 1 and 4) the first two steps remove it; where it is milky (rows 2 and 3) there is no cast to remove, and the steps only change the colours.

Water (open-sea style colour cast).

Glass. No method beat doing nothing on raindrops. The glass methods tried are DCP variants that change one assumption each: `dcp_near_global` (one almost constant t for the whole picture, because the film is on the pane and not at varying distance), `dcp_local_airlight` (glow colour estimated region by region, for an uneven film), and `flat_field` (divide out a slowly varying brightness).

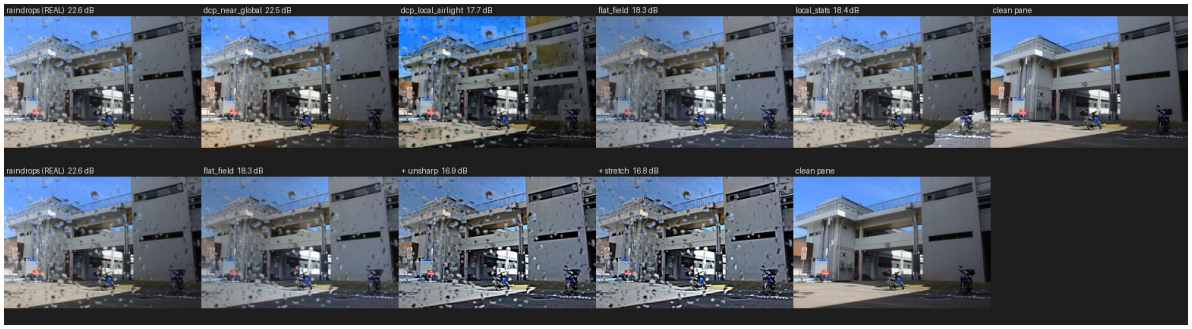


Figure 12: [REAL] Raindrops. Row 1: the glass-specific methods, each applied to the original, and the clean pane. Row 2: a chain `flat_field > unsharp > stretch`; each step lowers the PSNR (22.6, 18.3, 16.9, 16.8 dB). The methods change the colours and brightness of the whole picture but cannot recover what the drops hide, which is why the best chain for glass is the empty one.

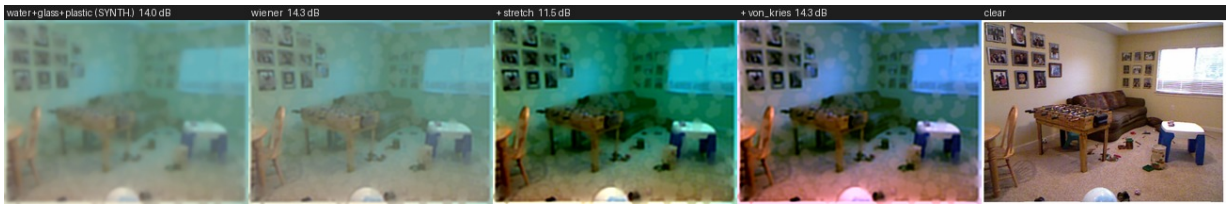


Figure 13: [SYNTHETIC] A picture seen through water, then glass with raindrops, then plastic, restored by the searched chain `wiener > stretch > von_kries`. The middle step makes the picture worse (14.3 to 11.5 dB) and the last step brings it back to 14.3: the chain ends where its first step already was, so the gain over doing nothing (14.0 dB) is small.

Stacked media.

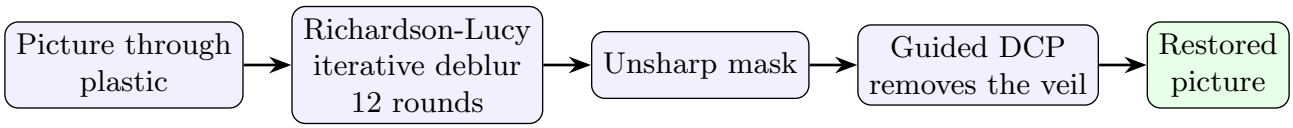


Figure 14: Best chain on synthetic plastic. The best single step is a plain linear contrast stretch.



Figure 15: [SYNTHETIC] Plastic, after each step of the chain (PSNR in each caption), the plain stretch, and the clear original. On this picture the chain gains +2.3 dB and the single stretch step +2.6 dB.

Plastic.

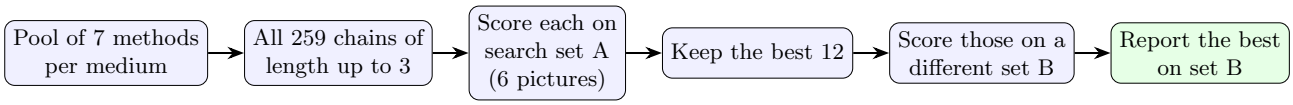


Figure 16: The step-order search. The best chain is chosen on one set and judged on another, so the result is not a lucky fit.

Medium	Best chain found	Assumption each step makes
Haze	hist_eq, CLAHE, guided DCP (synthetic); DCP, CLAHE (real)	Dark channel: some pixel in each patch is dark. CLAHE: contrast is lost locally, not just globally.
Water	red compensation, von Kries (+ unsharp)	Red is the colour lost; the average colour of the scene should be neutral.
Glass	none	Raindrops hide patches of the scene; no whole-picture step brings them back.
Plastic	Richardson-Lucy, unsharp, guided DCP (best single: stretch)	The blur is Gaussian and known; the rest is a dimming plus a milky veil.

Table 4: Summary of the pipelines. Scores are in Section 5–7.

How the best chains were found.

4.4 How we score

PSNR (in dB, higher is better) says how close the repaired picture is to the clean one pixel by pixel; **SSIM** (0 to 1) says how well the structure matches; **CIEDE2000** (lower is better) measures colour error. "*None*" means: do nothing to the spoiled picture. A method that scores below "none" makes things worse.

5 Results on made-up pictures [SYNTHETIC]

Here the clean answer is exact. We show the **calibrated** versions of haze and glass (settings tuned to look like the real thing; Section 7 explains why and what it changed). Water and plastic are the same in both versions.

5.1 Which method is best?

haze [SYNTHETIC]

method	PSNR dB	vs none	SSIM	CIEDE2000 ↓
hist_spec	14.49	+2.68	0.646	16.3
eq_hsi	14.01	+2.20	0.634	17.3
hist_eq	13.57	+1.76	0.627	17.9
hist_eq_rgb	13.14	+1.33	0.630	21.2
gamma_auto	12.90	+1.09	0.582	18.2
fusion	12.43	+0.62	0.592	20.9
none	11.81	+0.00	0.561	21.4

water [SYNTHETIC]

method	PSNR dB	vs none	SSIM	CIEDE2000 ↓
von_kries	16.59	+2.51	0.836	16.2
eq_avg_hist	15.42	+1.34	0.804	16.6
fusion	15.32	+1.24	0.789	18.8
hist_eq	15.31	+1.23	0.769	23.1
white_patch	15.22	+1.14	0.813	21.1
red_comp	15.21	+1.13	0.815	20.1
none	14.08	+0.00	0.790	22.6

glass (raindrops) [SYNTHETIC]

method	PSNR dB	vs none	SSIM	CIEDE2000 ↓
none	25.05	+0.00	0.861	2.3
freq_laplacian	24.92	-0.12	0.857	2.7
stretch	24.74	-0.30	0.854	3.1
unsharp	22.99	-2.06	0.787	4.8
dcp_near_global	22.57	-2.48	0.843	3.8
local_stats	21.15	-3.90	0.821	4.1

plastic [SYNTHETIC]

method	PSNR dB	vs none	SSIM	CIEDE2000 ↓
stretch	19.89	+4.12	0.685	10.4
hist_eq	17.64	+1.88	0.629	12.0
clahe	17.44	+1.68	0.638	11.6
eq_hsi	17.04	+1.28	0.627	13.4
retinex	16.81	+1.04	0.632	11.9
hist_spec	16.76	+0.99	0.586	13.4
none	15.76	+0.00	0.587	16.3

Table 5: [SYNTHETIC] The best methods per medium (PSNR averaged over the made-up pairs; "vs none" is the gain over doing nothing).

Plain reading of the table:

- **Haze:** histogram methods lead; DCP is barely better than nothing. (With our *original* hand-made haze, DCP was clearly best. That difference is the main story of Section 7.)
- **Water:** a colour correction (von Kries) wins, because water mostly shifts colours. (On real tank water nothing beats doing nothing; see Section 6.1.)
- **Glass:** nothing beats "none". Raindrops replace patches of the scene; no global operation can bring those patches back.
- **Plastic:** a plain contrast stretch wins; plastic is mostly a dimming, which a stretch undoes.

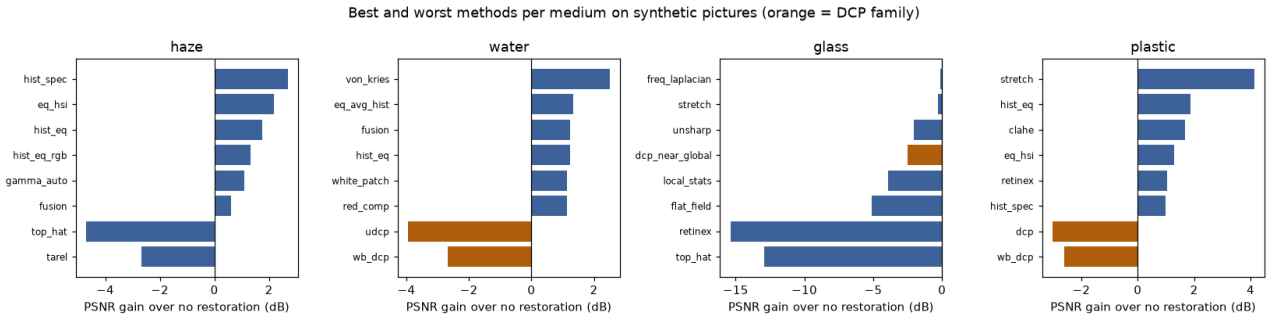


Figure 17: [SYNTHETIC] Gain over doing nothing, per method and medium (blue bars help, red hurt).

5.2 What the repairs look like

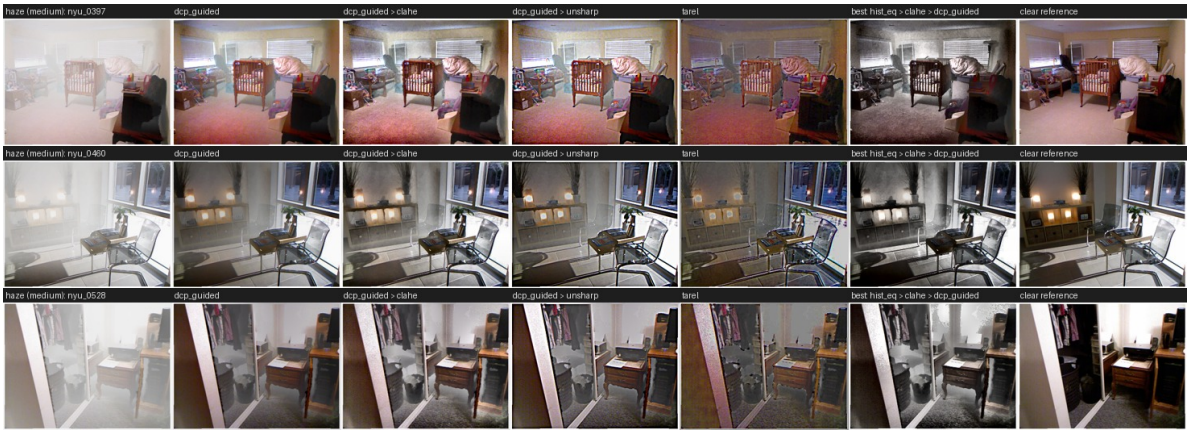


Figure 18: [SYNTHETIC] Haze. Each row is one picture: spoiled, several repairs, and the clean original on the right.

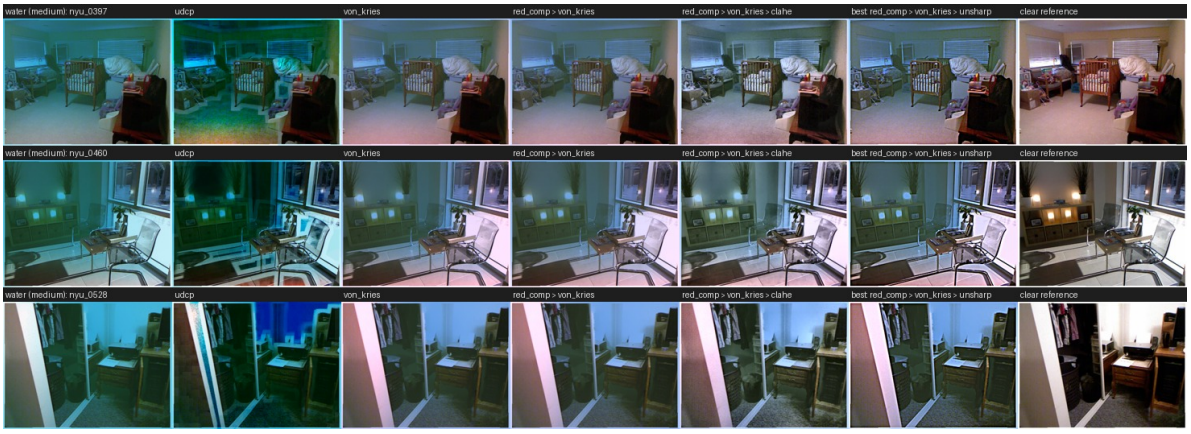


Figure 19: [SYNTHETIC] Water, same layout.

5.3 Does the order of steps matter?

We tried every chain of up to three steps from a short list, choosing the best on one set of pictures and checking it on a *different* set, so the result is not luck.

medium	best sequence (held-out)	PSNR	SSIM	DCP alone	no restoration
haze	hist_eq > clahe > dcp_guided	13.66	0.541	12.86	11.87
water	red_comp > von_kries	18.03	0.837	13.05	15.38
glass	none	25.80	0.854	19.78	25.80
plastic	richardson_lucy > unsharp > dcp_guided	17.53	0.667	15.27	16.03

Table 6: [SYNTHETIC] Best chain per medium on the held-out pictures, against DCP alone and doing nothing. `A > B` means run A first, then B on its output.

- **Yes, order matters**, by up to 3.1 dB between the best and worst order of the same steps.
- **Haze:** a contrast chain *before* DCP beats DCP alone (13.66 against 12.86 dB). So **something does beat DCP**, but only by a modest amount.
- **Water:** red-channel compensation then von Kries, +5.0 dB over DCP (18.03 against 13.05).
- **Plastic:** Richardson-Lucy deblur, sharpen, then DCP.

Post-processing is not free: histogram equalisation after gray-world helps haze (+3.8 dB) but hurts when applied after DCP (-1.5 dB).

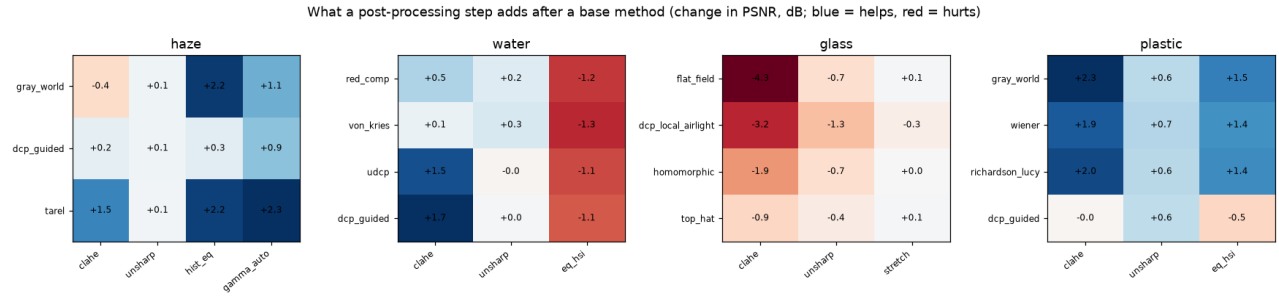


Figure 20: [SYNTHETIC] What an extra step adds after a base method (change in PSNR; blue helps, red hurts).

5.4 Stacked media

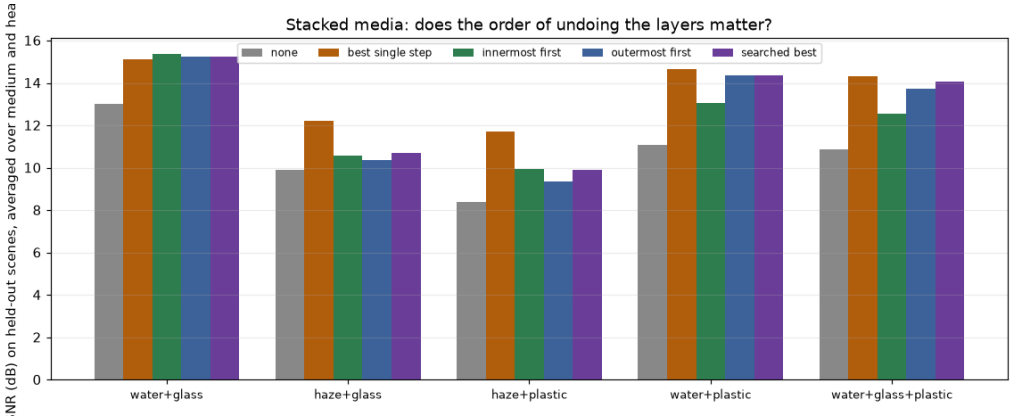


Figure 21: [SYNTHETIC] Two or three media stacked, with different orders of undoing them. With the calibrated glass the order matters little and not consistently.

6 Results on real pictures [REAL]

Now the real thing, where no model is involved.

real haze, 185 pairs [REAL]

method	PSNR dB	vs none	SSIM	CIEDE2000 ↓
fusion_dcp	14.76	+2.76	0.619	18.0
dcp_clahe	14.61	+2.61	0.630	18.6
dcp_guided	13.75	+1.75	0.536	21.0
fusion	13.62	+1.62	0.602	18.1
clahe	13.59	+1.59	0.635	18.9
dcp	13.58	+1.58	0.553	21.2
none	12.00	+0.00	0.492	21.9

real raindrops, 58 pairs [REAL]

method	PSNR dB	vs none	SSIM	CIEDE2000 ↓
none	24.02	+0.00	0.853	4.3
freq_laplacian	23.91	-0.11	0.841	4.5
dcp_near_global	22.13	-1.90	0.829	5.6
unsharp	21.80	-2.22	0.766	5.8
local_stats	21.48	-2.54	0.824	5.3
flat_field	20.49	-3.53	0.835	7.3

Table 7: [REAL] Best methods on real pictures: all 185 haze pairs and all 58 raindrop pairs. (A first version used only 36 pictures of each: the top two haze methods swapped places, a tie within 0.2 dB, and the raindrop result did not change.)

- **Real haze:** The two best are *fusion after DCP* (14.76 dB) and *DCP followed by CLAHE* (14.61 dB); doing nothing scores 12.00. DCP followed by CLAHE beats DCP alone by +0.86 dB (standard error 0.14; better on 138 of 185 pairs; SSIM +0.094). DCP alone leaves a dark picture with a blue cast; the contrast step brings brightness back.
- **Real raindrops:** again nothing helps ("none" scores highest, 24.02 dB; the best method, a frequency-domain Laplacian, is 0.11 dB lower and better on only 12 of 58 pairs).



Figure 22: [REAL] Haze. Hazy photo, three repairs, clean photo (right).

6.1 Real underwater pictures [REAL]

Two kinds of real underwater data.

TUB, paired tank photos. Objects sit in a water tank; milk is added in 10 steps while camera and objects stay fixed. The least turbid picture of each view is our reference. It is nearly clear (a few NTU, a unit of turbidity), not perfectly clear, so scores are against a nearly clear reference. Two cautions: the turbidity is milky scattering (brown-grey haze, much like fog), not the blue-green colour cast of the open sea; and the camera adjusts its exposure between steps, which PSNR counts as error.

water methods, 120 TUB pairs [REAL]

method	PSNR dB	vs none	SSIM	CIEDE2000 ↓
red_comp	18.82	+0.10	0.740	10.0
gamma_auto	18.78	+0.06	0.741	9.6
none	18.72	+0.00	0.740	10.0
fusion	17.80	-0.92	0.727	15.9

haze methods (DCP and others), same pairs [REAL]

method	PSNR dB	vs none	SSIM	CIEDE2000 ↓
unsharp	18.82	+0.10	0.741	10.0
gamma_auto	18.78	+0.06	0.741	9.6
none	18.72	+0.00	0.740	10.0
fusion_dcp	18.35	-0.37	0.683	9.9

Table 8: [REAL] Repair of real turbid tank water, averaged over 120 pairs (40 per turbidity band).



Figure 23: [REAL] Paired tank water. Rows: mild, medium, heavy turbidity. Left: turbid picture; middle: repairs; right: the least turbid picture of the same view.

Reading. Nothing beats doing nothing by more than 0.1 dB. DCP makes things much worse (11.45 dB against 18.72 for nothing): it darkens the picture and shifts the colours. CLAHE scores 15.83 and DCP followed by CLAHE 13.98, both worse than doing nothing. In the pictures the repairs do bring out colour and contrast (DCP, CLAHE) but they also darken the scene and amplify the noise, and none of them returns the detail that is lost: at heavy turbidity almost nothing is left to recover.

UIEB, ocean and reef photos. No clean twin exists, so we can only look at them and use reference-free scores.

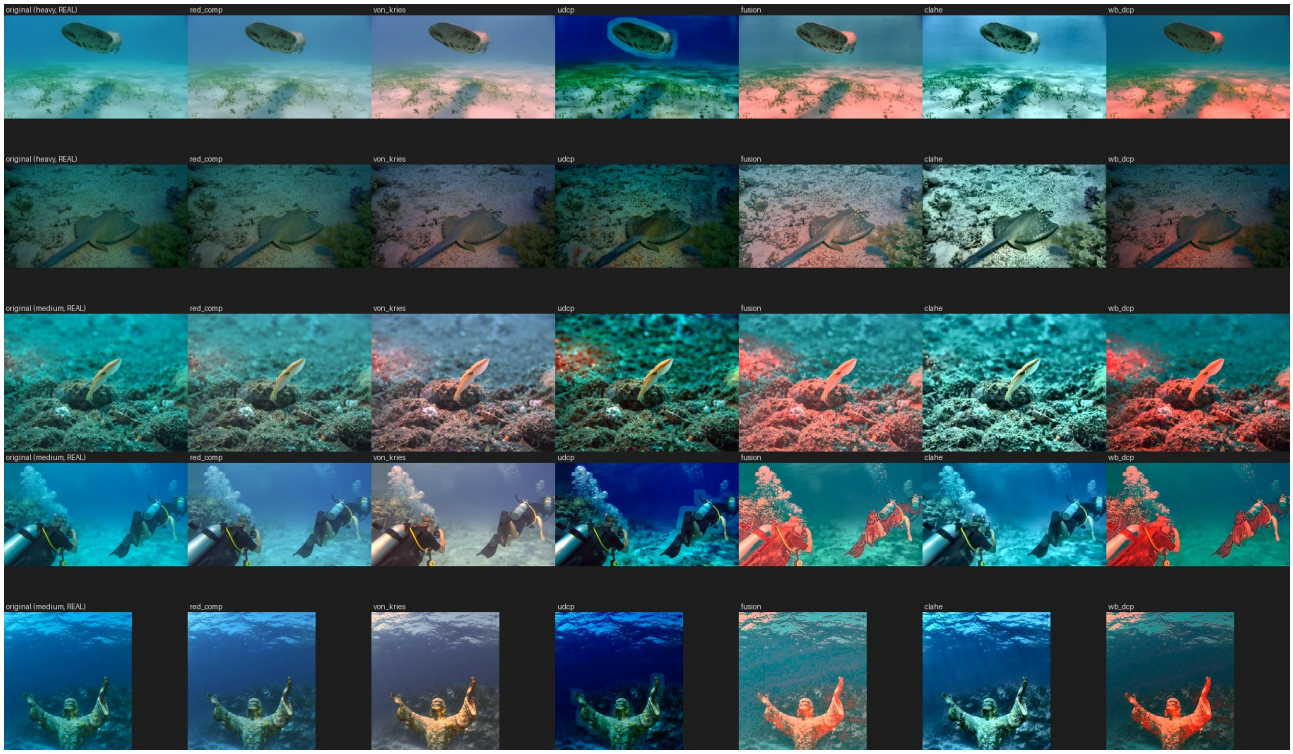


Figure 24: [REAL] Ocean and reef photos (no clean reference). Left: original; then six repairs. Colour correction (red compensation, von Kries) brings back warm colours; `wb_dcp` and `fusion` over-correct and turn the seabed red; UDCP turns the scene dark blue.

On these photos the problem is mostly *colour*, and the colour methods work visibly better than DCP-style methods, in line with the synthetic water results. We cannot put a number on it, because reference-free scores disagree with each other (UCIQE prefers UDCP and stretch, while the remaining colour cast prefers equalised-average-histogram, fusion and von Kries; see the project log, finding F55).

7 Can made-up pictures stand in for real ones?

We compared **[SYNTHETIC]** with **[REAL]** in seven ways, from the simplest to the most demanding.

7.1 Test 1: do they look statistically alike?

For each spoiled picture we measure five simple things against its clean twin: how much the darkest values rise (*dark shift*), how much contrast is lost (*contrast ratio*, 1 = unchanged), how much brighter it is, how much detail remains (*sharpness ratio*), and the share of the picture that changed.

medium	level	pictures	dark shift	contrast ratio	brightness shift	sharpness ratio
haze	mild	real	0.20	0.70	0.10	0.40
haze	mild	synthetic, original	0.21	0.81	0.14	0.72
haze	mild	synthetic, calibrated	0.23	0.56	0.09	0.47
haze	medium	real	0.30	0.57	0.17	0.29
haze	medium	synthetic, original	0.37	0.72	0.25	0.51
haze	medium	synthetic, calibrated	0.32	0.49	0.15	0.33
haze	heavy	real	0.42	0.40	0.25	0.13
haze	heavy	synthetic, original	0.51	0.65	0.34	0.34
haze	heavy	synthetic, calibrated	0.44	0.40	0.23	0.17
glass	mild	real	-0.00	0.92	-0.00	1.01
glass	mild	synthetic, original	0.07	0.89	0.05	0.88
glass	mild	synthetic, calibrated	0.01	0.96	0.01	0.96
glass	medium	real	0.01	0.88	0.00	0.92
glass	medium	synthetic, original	0.17	0.75	0.11	0.73
glass	medium	synthetic, calibrated	0.01	0.94	0.01	0.88
glass	heavy	real	0.02	0.85	0.01	0.88
glass	heavy	synthetic, original	0.29	0.57	0.18	0.55
glass	heavy	synthetic, calibrated	0.03	0.92	0.01	0.80

Table 9: **[REAL vs SYNTHETIC]** How much each degradation changes the picture. The closer a synthetic row is to the real row above it, the better the match.

Reading. Our original haze was too bright and too sharp (contrast ratio 0.65–0.81 against 0.40–0.70 in real fog; sharpness 0.34–0.72 against 0.13–0.40). The calibrated haze is much closer (sharpness 0.17 against 0.13 for heavy, contrast 0.40 against 0.40). Fitting the haze model to real pairs gave a glow of brightness $A \approx 0.60$–0.70, not our original 0.92. Real raindrops barely change brightness or contrast (ratio near 0.9), so a veil model is wrong for them; droplets-only fits.

7.2 Test 2: do the same methods win?

For every method we plot its score on synthetic pictures against its score on real ones. If the models were perfect, the dots would sit on the dashed line, and a high-scoring method on one side would be high-scoring on the other.

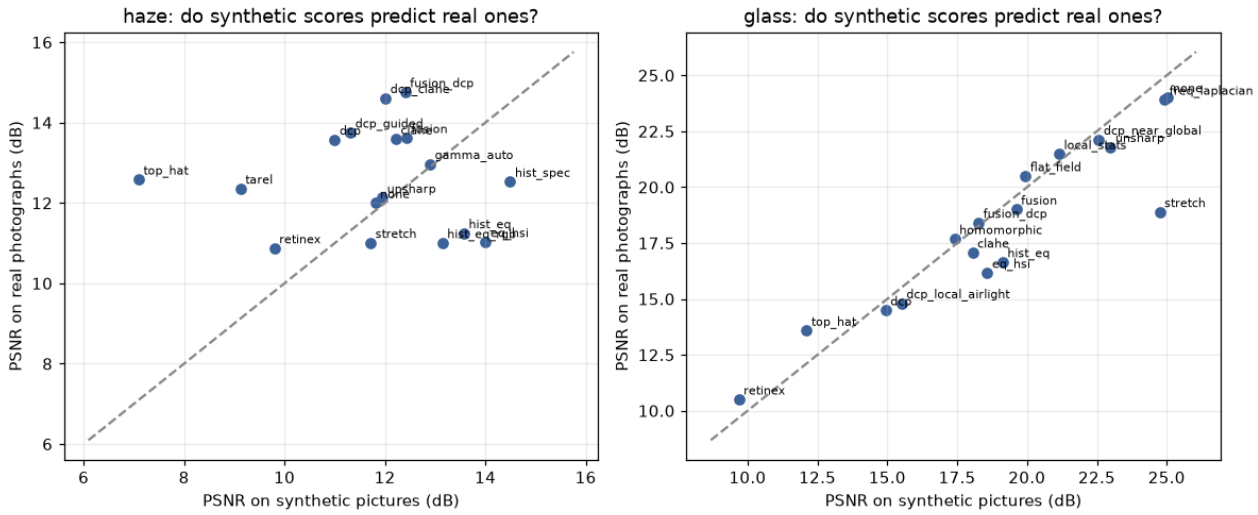


Figure 25: [REAL vs SYNTHETIC] Each dot is one method. Left: haze (synthetic is the calibrated version). Right: raindrops on glass.

- **Glass:** the dots follow the line closely. Methods rank almost the same (Spearman correlation 0.92 on 12 scenes, 0.85 on 90). The one clear outlier is *stretch*, which scores 24.7 dB on synthetic and 18.6 dB on real raindrops.
- **Haze:** the dots are scattered (correlation −0.04 on the 12 scenes of this figure; with 90 scenes it is +0.30, see Test 7). Histogram methods (`hist_spec`, `eq_hsi`) score well on synthetic and poorly on real; the DCP-plus-CLAHE family scores best on real and middling on synthetic.

medium	presets	best single	PSNR	guided DCP	none	best chain (held-out)	PSNR	rank agreement
haze	original	dcp_guided	15.63	15.63	10.52	dcp_guided	15.26	+0.57
haze	calibrated	hist_spec	14.49	11.31	11.81	hist_eq > clahe > dcp_guided	13.66	-0.04
glass	original	stretch	28.52	–	18.51	stretch	26.87	+0.53
glass	calibrated	none	25.05	–	25.05	none	25.80	+0.92

Table 10: [REAL vs SYNTHETIC] Original hand-chosen presets against calibrated ones. Rank agreement is the correlation between how methods rank on synthetic and on real pictures (1 = identical, 0 = unrelated).

7.3 Test 3: what did calibration change?

We tuned the settings of haze and glass to look like the real thing, then reran the experiments. The old results are kept in `results_original_presets/`.

- **With the original haze,** DCP looked excellent (+5.1 dB over nothing). On calibrated haze and on real haze it is weak. *The original settings flattered DCP.*
- **With the original glass,** a plain stretch gained +10 dB. On raindrops (calibrated and real), nothing helps. *The original settings flattered stretch.*
- **Glass agreement rose** from 0.53 to 0.92.
- **Haze agreement got worse with calibration.** Against all 185 real pairs, the original haze ranks the methods at +0.57 and the calibrated haze at −0.04 (12 scenes each); Test 7 repeats this on 90 scenes (+0.57 against +0.30). Calibration matched the *statistics* in Test 1, not the *ranking*.

7.4 Test 4: remove the scene as an excuse

In Tests 2 and 3 the synthetic pictures show indoor rooms and the real haze shows outdoor gardens, so a mismatch could simply be the scenery. Test 4 takes each real pair, degrades its *own clean picture* with our model, and compares the real spoiled picture with its synthetic twin.

medium	pictures	Spearman	Kendall	gain corr.	best on real	best on twin	$\lvert\Delta\text{PSNR}\rvert$ dB
haze	91	+0.81	+0.66	+0.54	fusion_dcp	dcp	1.22
glass	58	+0.99	+0.97	+0.93	none	none	1.28

Table 11: **[REAL vs SYNTHETIC]** Same scene, real degradation against synthetic twin. The model for haze uses the strength fitted on the other half of the same dataset.

Now the rankings agree for **haze (+0.81)** and **glass (+0.99)**, and a method that helps a real picture also tends to help its twin (per-picture correlation 0.54 for haze, 0.93 for glass). So the model itself is reasonable for haze and good for glass when scene and haze strength are the same as in the real picture. Test 7 shows that this does not carry over to other scenes: with other scenes (indoor or outdoor) the ranking agreement is only +0.1 to +0.5.



Figure 26: **[REAL vs SYNTHETIC]** Clean picture, the real spoiled picture, and our synthetic twin of it.

7.5 Test 5: water, now with real pairs

The real tank pairs let us do the same ranking test for water.

turbidity band	pairs	none (PSNR)	best method on real	rank agreement with synthetic
all	120	18.72	red_comp	+0.41
mild	40	21.17	red_comp	+0.43
medium	40	18.01	gamma_auto	+0.40
heavy	40	16.98	gamma_auto	+0.32

Table 12: **[REAL vs SYNTHETIC]** Do the 17 water methods rank alike on real tank water and on synthetic water? Rank agreement is the correlation between the two rankings.

Agreement is moderate (+0.3 to +0.4) and falls with turbidity. The real winners are different from the synthetic ones: on synthetic water von Kries wins by a large margin; on real tank water nothing beats doing nothing. One plausible reason: milky tank water is scattering, close to the haze model, not the colour absorption our water model emphasises. Test 6 tests exactly that.

7.6 Test 6: synthetic water with milky scattering [SYNTHETIC]

We built a second synthetic water, *milky water*, that matches the tank rather than the open sea. Open-sea water absorbs red first. Milk in a tank mainly *scatters* all colours alike, so the picture fades towards a light brown-grey, like fog. Its equation is the haze equation with three changes: the same β for all colours, the glow colour measured on the most turbid tank pictures (about 0.55, 0.52, 0.41 in red, green, blue), and an optional blur, because forward scattering smears the scene (in the tank, detail was lost faster than contrast). We chose the density β and the blur for each turbidity band so that the five signature numbers of Test 1 match the real tank pairs.

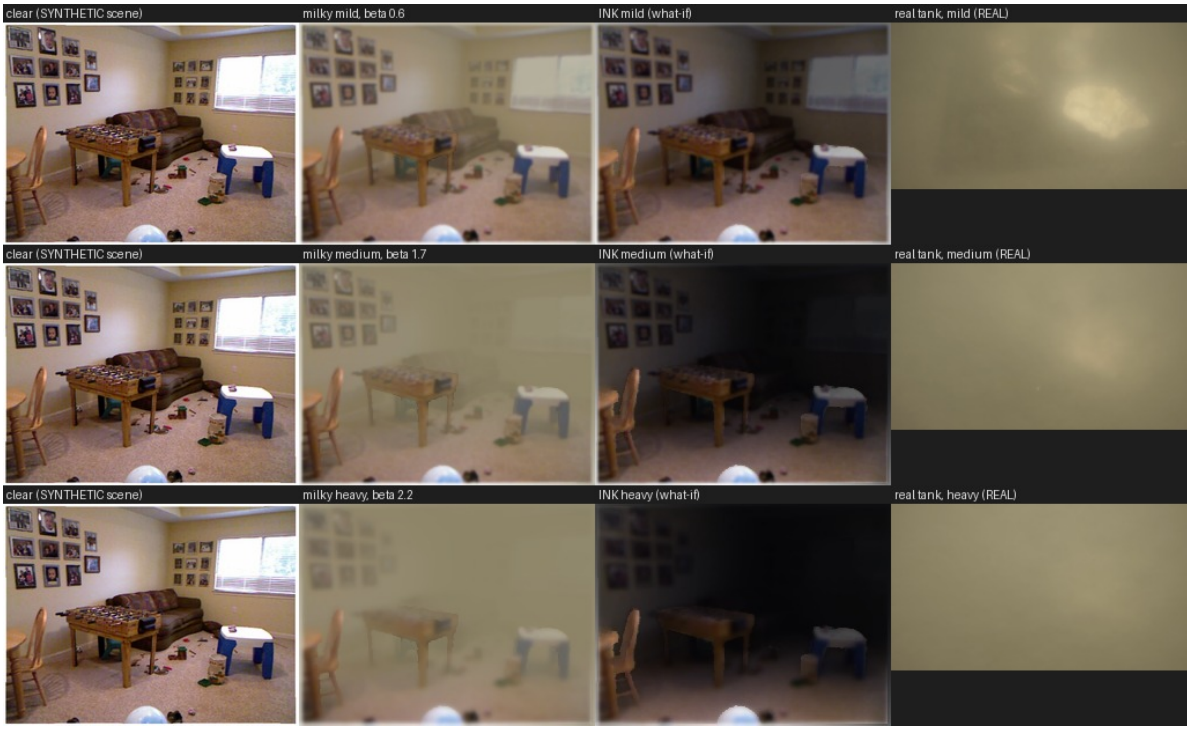


Figure 27: **[REAL vs SYNTHETIC]** Left to right: a clear synthetic scene, the same scene with calibrated milky water, the same density with a black cloud (the *ink* what-if, next subsection), and a real tank picture of the same turbidity band (a different scene). Rows: mild, medium, heavy.

band	pictures	beta	blur	contrast ratio	brightness shift	sharpness ratio
mild	real tank	–	–	0.60	0.05	0.32
mild	synthetic milky	0.6	0.004	0.65	0.02	0.40
medium	real tank	–	–	0.40	0.06	0.14
medium	synthetic milky	1.7	0.004	0.45	0.05	0.24
heavy	real tank	–	–	0.31	0.07	0.10
heavy	synthetic milky	2.2	0.008	0.39	0.06	0.15

Table 13: [REAL vs SYNTHETIC] How much each degradation changes the picture, real tank against calibrated synthetic milky water. Contrast and sharpness ratio: 1 = unchanged.

The match is only fair (the signature distance stays around 0.65 to 0.72 real spreads): our milky water fades the picture evenly, whereas the real tank had bright patches from the lamp and uneven light. Even so, the test result moved in the right direction.

method list	band	original water	milky water	best on real	best on milky
water methods	all	0.41	0.41	red_comp	hist_eq
water methods	mild	0.43	0.46	red_comp	stretch
water methods	medium	0.4	0.3	gamma_auto	hist_eq
water methods	heavy	0.32	0.41	gamma_auto	hist_spec
haze methods	all	–	0.43	unsharp	unsharp
haze methods	mild	–	0.6	unsharp	unsharp
haze methods	medium	–	0.19	gamma_auto	hist_eq
haze methods	heavy	–	0.22	gamma_auto	hist_spec

Table 14: [REAL vs SYNTHETIC] Rank agreement (correlation) between the real tank result and each synthetic water. "Water methods" are the 17 colour-oriented methods, "haze methods" the 17 DCP-family methods. The original water was only run with the water methods.

method	PSNR dB	vs none
unsharp	15.42	+0.45
hist_eq	15.19	+0.22
hist_spec	15.12	+0.15
shades_of_gray	15.02	+0.05
none	14.97	+0.00

Table 15: [SYNTHETIC] Milky synthetic water: the best methods barely beat doing nothing (calibrated, 12 scenes, three densities).

Reading.

- **The verdict now agrees.** On the original synthetic water the colour methods won by a lot (von Kries +2.5 dB over nothing). On milky synthetic water, as on the real tank, the best method beats doing nothing by under 0.5 dB, and DCP-style methods hurt (guided DCP 12.8 against 15.0 for nothing; on the real tank 11.5 against 18.7). The size of the DCP loss is smaller on the synthetic water than on the real one.
- **The full ranking only improves a little.** Overall agreement stays at +0.41 for the water methods and reaches +0.43 for the haze methods (+0.60 in the mild band, about +0.2 in the medium and heavy bands). When nearly every method is within a dB of doing nothing, the ranking is decided by small differences and is hard to reproduce.

7.7 Does milky water look like squid or octopus ink? [SYNTHETIC]

In what they share. A cloud of ink in water is also a cloud of particles between the scene and the camera: it lets a fraction t of the scene light through and adds light of its own. The same equation $I = tJ + A(1 - t)$ describes it, and it spoils the picture in the same way, by hiding detail.

In what differs. Milk particles scatter light and absorb almost none, so the cloud glows (large A, light colour). Ink (mostly the dark pigment melanin) absorbs strongly, so the cloud is dark (A close to zero) and the scene fades towards *black* instead of white. Real ink is also a patchy plume that spreads and dissolves within seconds, not an even cloud, so a spatially uniform model is a rough picture of it at best.

To see what that difference does to a repair, we ran the same density as the milky water but with a near-black cloud ($A \approx 0.02$). There is no real ink data, so this is a what-if, not a calibration.

method	PSNR dB	vs none
fusion	13.00	+2.12
white_patch	12.88	+1.99
gamma_auto	12.63	+1.74
clahe	12.54	+1.65
none	10.89	+0.00

Table 16: [SYNTHETIC] Ink what-if (black cloud, same densities as the milky water, 12 scenes). Compare with the milky table above.

Here repairs *do* help: the best gains +2.1 dB over nothing (fusion), and CLAHE, white-patch and automatic gamma gain 1.7 to 2.0 dB, because a dark cloud mostly lowers brightness, which brightening methods undo. DCP still hurts (9.3 against 10.9 dB for nothing). A likely reason, not tested: DCP assumes a bright glow and the glow here is black, so it estimates the wrong glow. So milky and inky water share an equation but not a remedy.

7.8 Test 7: more scenes, and outdoor scenes

Two doubts remained: the haze numbers of Tests 2 and 3 use only 12 indoor rooms, and the real haze is outdoors. We repeated the haze comparison with more scenes and a different kind of scene:

- **90 indoor rooms** (NYU), both the original and the calibrated haze;
- **100 outdoor pictures** from RESIDE-SOTS, which comes with clear scenes and haze made by its own authors (so it tests the scene type but not our haze model).

pictures	n	Spearman vs real haze	best method	PSNR	guided DCP	none
indoor rooms, original haze (synthetic)	90	+0.57	dcp_guided	15.63	15.63	10.16
indoor rooms, calibrated haze (synthetic)	90	+0.30	hist_spec	13.46	12.42	11.28
indoor rooms, refined haze: per-colour t and glow, blur (synthetic)	90	+0.28	hist_spec	13.14	12.16	11.21
indoor rooms, uniform fog, no depth (synthetic)	90	-0.12	stretch	47.72	18.68	12.88
outdoor, RESIDE-made haze (synthetic)	100	+0.09	stretch	23.83	18.68	16.01

Table 17: [SYNTHETIC] Rank agreement with real haze over the 17 haze methods, and the winners. All pictures here are synthetic; "real haze" is the reference of all 185 real pairs (best: fusion after DCP 14.76 dB; DCP then CLAHE 14.61; guided DCP 13.75; none 12.00).

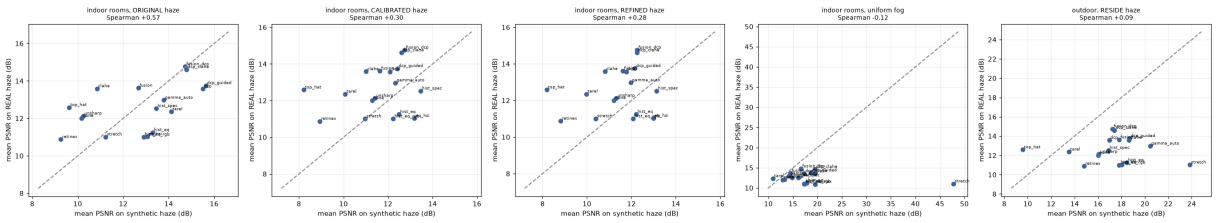


Figure 28: [REAL vs SYNTHETIC] Synthetic against real PSNR per method, for the three sets of synthetic haze.

- **Twelve scenes are noisy for the calibrated haze.** Drawing 12-scene subsets from the 90 gives rank agreements from −0.06 to +0.55 (90% of draws, calibrated haze) and +0.51 to +0.63 (original haze).
- **Calibration made the ranking of haze worse.** On 90 scenes the original haze agrees at +0.57 and the calibrated at +0.30; resampling both the scenes and the real pairs, the original is better by +0.30 (90% interval +0.13 to +0.50, positive in every one of 2000 draws; this covers picture sampling only). The calibrated haze changes the winner (histogram specification instead of DCP), but the real winners are fusion after DCP and DCP then CLAHE, both DCP-family methods, which the *original* haze favours. Calibration matched the picture statistics, not the method ranking.
- **Outdoor scenes did not make it better.** The RESIDE outdoor haze agrees at only +0.09 and its winner is a plain stretch (23.8 dB). Its haze is much lighter (a global stretch undoes it), so this set differs from real fog in haze model as well as scene, which means it does not settle the indoor-versus-outdoor question. A depth-based outdoor test (DIODE) was not possible: the download ran at 12 to 54 kilobytes per second (over 13 hours) and was stopped.

7.9 How stable are these agreement numbers?

The agreement numbers above come from small samples. To see how much they could move, we resampled both sides (the synthetic scenes and the real pairs) 2000 times with replacement and recorded the spread, and we re-made the synthetic side with 90 scenes instead of 12.

comparison	scenes / pairs	first estimate (90% interval)	with 90 synthetic scenes (90% interval)
glass against real raindrops	12 / 58	+0.92 (+0.83 to +0.95)	+0.85 (+0.80 to +0.90)
water (original) against real tank	12 / 120	+0.41 (+0.29 to +0.50)	+0.32 (+0.27 to +0.39)
water (milky) against real tank	12 / 120	+0.41 (+0.21 to +0.60)	+0.45 (+0.33 to +0.58)
same-scene twins, haze	91 / 91	+0.81 (+0.67 to +0.84)	–
same-scene twins, glass	58 / 58	+1.00 (+0.97 to +1.00)	–

Table 18: **[REAL vs SYNTHETIC]** Rank agreement between synthetic and real, with the 90% bootstrap interval (it accounts for the choice of pictures only, not for other uncertainty such as the preset settings). The real glass sample is all 58 pairs. The same-scene rows resample the shared pictures.

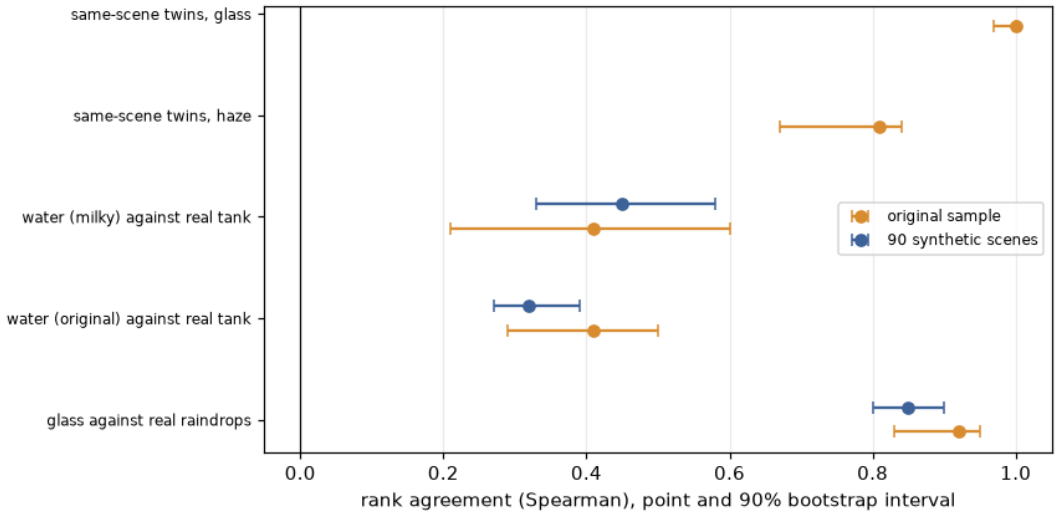


Figure 29: **[REAL vs SYNTHETIC]** The same numbers as intervals.

- **Glass is stable and strong:** +0.85 with 90 scenes (+0.79 to +0.89). The earlier 0.92 came from only 12 scenes; the true value is a little lower, still a strong agreement.
- **Original synthetic water against the real tank is weak and stable:** about +0.3 (+0.27 to +0.39 with 90 scenes).
- **Milky synthetic water is a modest improvement:** +0.45 (+0.33 to +0.59) against +0.32 for the original water. The intervals overlap a little, so the evidence is suggestive, not firm.
- **The same-scene twins are the most reliable:** haze +0.81 (+0.67 to +0.84), glass +1.00. Where the scene and haze strength match, the synthetic model ranks the methods like the real picture does; the agreement fades when the scenes differ.

7.10 So what can we trust?

Question	Verdict
Haze: best method	Real and calibrated synthetic agree that DCP alone is weak and a contrast step helps.
Haze: full ranking	Moderate to weak: +0.30 (calibrated) to +0.57 (original) on 90 indoor scenes, +0.09 on outdoor scenes with RESIDE's haze, +0.81 on the *same scenes* as the real photos. Different scenes, and even different haze models, move it a lot.
Glass (raindrops)	Synthetic is a good stand-in: +0.85 (+0.79 to +0.89) on 90 scenes, up to +1.00 on the same scenes.
Water	**Partly checked.** The original (open-sea style) synthetic water disagrees with real tank pairs (+0.3 to +0.4, and a different winner). Milky synthetic water agrees on the verdict (nothing helps much, DCP hurts) but only modestly on the ranking (+0.4). On real ocean photos colour methods look best by eye. Ink: no real data.
Plastic	**Unchecked**: no real data exists. Treat as model results.
Original presets	Misleading for DCP on haze and stretch on glass; kept only for comparison.

8 Global or local? Where in the picture should a method act?

Haze thickens with distance, raindrops cover only parts, plastic is the same everywhere. So should a method decide once for the whole picture (*global*) or separately for each part (*local*)? We ran the same method both ways:

- **tiles:** cut the picture into a 2×2, 4×4 or 8×8 grid, run the method on each tile, blend the tiles back smoothly;
- **regions found from the picture** (`dark3`, `sharp2`): group pixels into three levels of "haze thickness" (from the dark channel) or into blurred and sharp areas, and run the method once per group;
- **regions from the true depth** (`depth3`): only possible on [SYNTHETIC] pictures, so it shows the best a perfect region-finder could give.

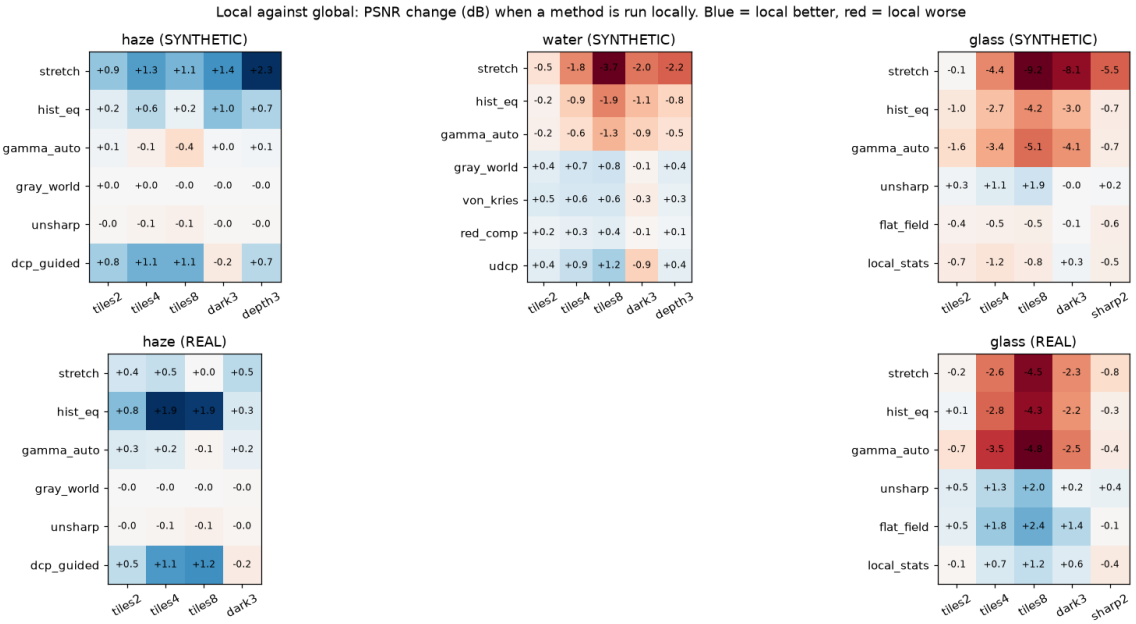


Figure 30: [REAL vs SYNTHETIC] Change in PSNR (dB) when a method runs locally instead of globally. Blue: local is better; red: worse. Top row [SYNTHETIC], bottom row [REAL]. Water has no real pairs.

What we see.

- **Haze (local helps, on both kinds of pictures).** On real haze, histogram equalisation on 4×4 tiles gains +1.95 dB (better on every picture) and turns a method that was worse than doing nothing (10.8 against 11.6 dB) into a useful one (12.7). Guided DCP gains +1.2 dB on 8×8 tiles. On synthetic haze, stretch gains +2.3 dB when given the true depth regions. Depth is the useful information: it is what makes the haze differ from place to place.
- **Water (small gains, synthetic only).** Colour methods gain a little locally (gray-world +0.8, von Kries +0.6), contrast methods lose.
- **Glass (local hurts).** Stretch and histogram equalisation lose up to 9 dB when run per tile, on synthetic and real raindrops. A tile that contains a drop is stretched differently from its neighbour, which creates patchwork. The apparent gains of unsharp, flat-field and local-statistics are *not* locality: these methods measure their scale as a share of the picture size, so on a small tile they act more weakly and drift towards doing nothing, which still beats every variant (24.5 dB on real).

8.1 How many tiles?

We repeated the comparison with every grid from 2×2 to 12×12 tiles. A grid with n tiles per side cuts the picture into $n \times n$ tiles (4×4 is 16 tiles).

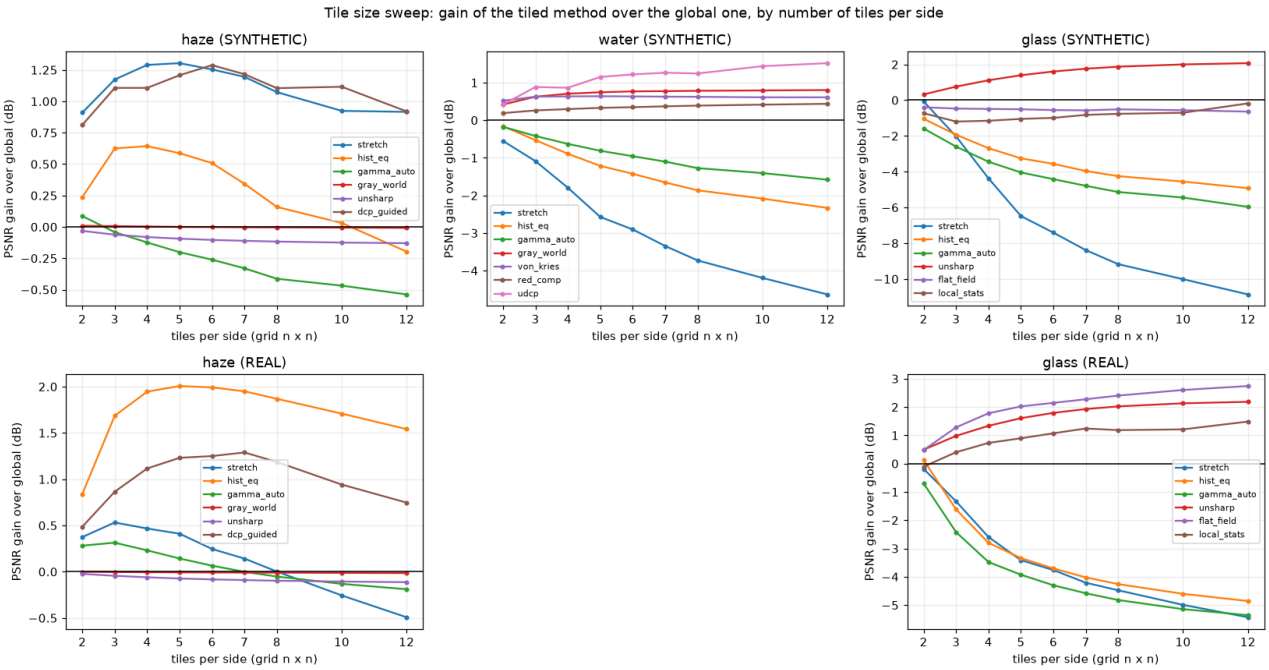


Figure 31: [REAL vs SYNTHETIC] Gain over the global method for each grid size, one line per method. Top row [SYNTHETIC], bottom row [REAL].

Pictures	Medium	2	3	4	5	6	7	8	10	12
[REAL]	haze	+0.33	+0.56	**+0.62**	**+0.62**	+0.58	+0.55	+0.48	+0.36	+0.25
[SYNTHETIC]	haze	+0.34	+0.47	+0.47	+0.47	+0.45	+0.39	+0.30	+0.25	+0.16
[SYNTHETIC]	water	**+0.09**	+0.05	−0.12	−0.25	−0.33	−0.44	−0.55	−0.63	−0.74
[REAL]	glass	**+0.02**	−0.45	−0.84	−1.03	−1.13	−1.23	−1.32	−1.47	−1.54
[SYNTHETIC]	glass	**−0.59**	−1.24	−1.84	−2.32	−2.56	−2.80	−2.99	−3.21	−3.42

Table 19: Mean PSNR gain (dB) over the global method, averaged over the methods tried per medium, by tiles per side. Best grid in bold.

- **Haze: about 4×4 to 6×6 is best, then the gain falls.** By method: histogram equalisation peaks at 5×5 on real haze (+2.0, but the curve is flat from 4 to 8); DCP peaks later, at 6×6 or 7×7 (+1.3), because its own patch size is measured relative to the tile; gray-world and unsharp do not change.

- **The best size is not the same across media.** In water the colour methods (gray-world, red compensation, UDCP) keep gaining up to 12×12, while stretch and histogram equalisation lose more with every extra tile. In glass the less tiling the better: stretch falls from −0.1 dB at 2×2 to −10.9 dB at 12×12 on synthetic raindrops.
- **Why:** haze changes smoothly with distance, so medium-sized tiles each see roughly constant haze but still enough picture to decide sensibly. Small tiles see too little content. Raindrops are patches, and a tile that contains one is stretched differently from its neighbour.

Unsharp, flat-field and local-statistics seem to improve on glass with more tiles. That is not locality: on a small tile they act more weakly and drift towards doing nothing.

Caution. Findings come from one set of regions (three levels of dark channel) and a few tile sizes; PSNR penalises local re-stretching that shifts brightness; and the table of best local variants in the log is a maximum over several tries, so it is optimistic (the figure shows every try).

9 Why does it work this way?

9.1 How long should a chain be? A mathematical check, then a test

We searched chains of up to 3 steps. Would 4 or 5 steps help? Before searching, we asked what the mathematics allows.

What the equations say. Haze, water, a film on glass and plastic each damage a picture by (optionally) a blur and then a per-colour *affine* map, $I = tJ + A(1-t)$. Affine maps compose into one affine map, so a single step (a stretch, a white balance) undoes any number of such layers, and the blur needs one more step. The damage is therefore invertible in *one or two steps*. Two further facts give ceilings:

- **A ceiling for chains of whole-picture brightness steps.** Stretch, gamma, histogram equalisation, white balance and log are all a function of the pixel value, so a chain of any number of them is still one such function per colour. The best possible function (the *oracle LUT*: for each input value, the average true value of the pixels that have it) is a ceiling for every such chain, however long.
- **A ceiling for any chain on invertible damage.** Inverting with the true t and A is exact except for the rounding of the saved picture to 8 bits (a grey level of $1/255$), which the inversion multiplies by $1/t$. Its predicted PSNR is $-10\log_{10}\left(\overline{1/t^2}/(12\cdot 255^2)\right)$.

medium	none	1 step	2 steps	3 steps	oracle LUT	oracle affine	true inverse	predicted inverse
haze	11.86	12.58	13.50	13.52	19.79	15.00	30.89	18.70
water	15.36	17.82	18.00	17.96	22.48	19.88	46.11	45.85
glass	25.88	24.57	22.56	19.44	26.77	26.37	–	–
plastic	16.01	18.61	16.94	17.23	24.84	24.40	30.82	54.33

Table 20: [SYNTHETIC] Held-out PSNR (18 pictures): doing nothing; the best chain of 1, 2 and 3 steps (chosen on other pictures); the two ceilings; and the predicted PSNR of the true inverse. No inverse exists for raindrops.

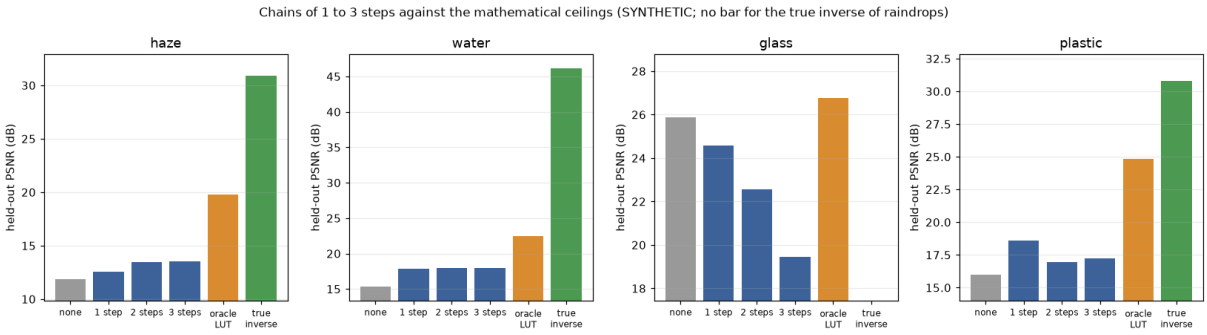


Figure 32: [SYNTHETIC] The best chains against the ceilings.

What we checked.

- **The formula holds where the damage is affine.** Water: predicted 45.9 dB, measured 46.1 dB. For haze the measured value is higher than predicted (30.9 against 18.7) because the formula ignores that pictures are clipped to the valid range, so the error cannot grow without limit. For plastic the formula predicts 54 dB but the measured inverse is 30.8 dB: the missing 24 dB is the cost of undoing the blur, which the formula does not include.
- **No chain reaches the ceilings, and the gap is large** (haze 13.5 against 30.9 dB for the true inverse; water 18.0 against 46.1; plastic 17.2 against 30.8). The limit is not the number of steps, it is that the chain must *guess* t, A, the tint and the blur from the damaged picture alone. Even the best single affine map (which needs the clear picture) is far above the found chains for plastic (24.4 against 17.2 dB).
- **So extra steps should add nothing once the damage is undone.** The prediction: the gain of a step beyond the 2nd or 3rd is zero within noise.

The test. We searched chains of 1 to 3 steps completely and chains of 4 and 5 steps by extending the best 40 chains of the previous length (a beam search), choosing the best chain of each length on one set of scenes (A). A second set (B) chose the length (the shortest within one standard error of the best), and a third set (C, 36 pictures, never used to choose anything) reports the result.

medium	set	0	1	2	3	4	5	length chosen on B
haze	B	11.87	12.65	13.55	13.66	13.23	13.33	2
haze	C	11.49	13.08	14.94	15.29	15.14	15.26	
water	B	15.38	17.85	18.03	17.98	16.91	15.85	2
water	C	14.30	16.88	17.06	17.13	16.20	15.04	
glass	B	25.80	24.52	22.57	19.43	16.02	14.32	1
glass	C	25.04	24.96	22.70	19.20	16.80	14.74	
plastic	B	16.04	18.63	17.03	17.25	17.19	14.76	1
plastic	C	15.73	17.89	18.00	18.06	17.67	15.92	

Table 21: [SYNTHETIC] Held-out PSNR of the best chain of each length (0 = doing nothing). B chose the length; C was never used.

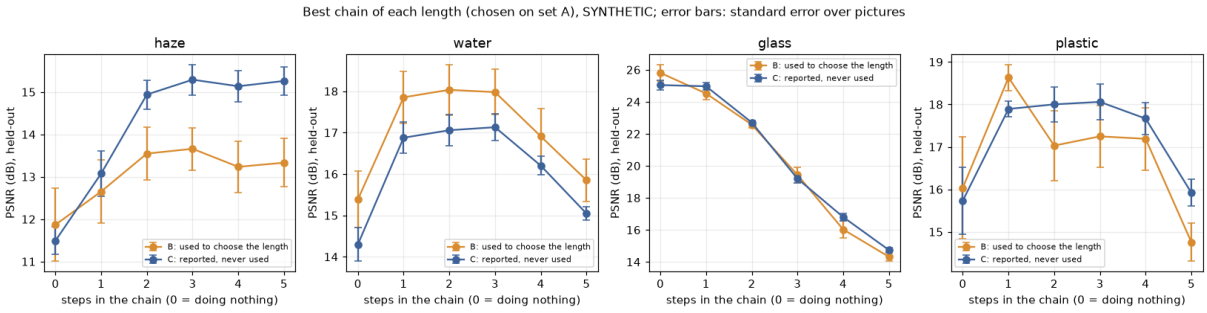


Figure 33: [SYNTHETIC] Best chain of each length against the number of steps. Error bars: standard error over pictures.

- **The prediction held.** The gain of the 4th step is -0.15 ± 0.06 dB (haze), -0.9 ± 0.2 (water), -2.4 ± 0.2 (glass), -0.4 ± 0.1 (plastic) on set C, and the 5th step gains $+0.13 \pm 0.13$ at best (haze) and loses 1 to 2 dB elsewhere. Among the chains searched, those of 4 and 5 steps were not even best on the search set.
- **Where the curve peaks.** Haze: 2 to 3 steps (set C: 3 steps $+0.35 \pm 0.14$ over 2; set B, with half the pictures, chose 2). Water: 2 steps. Plastic: 1 to 3 steps are tied within noise. Glass: every step lowers the score, and no chain beats doing nothing.
- **More steps usually hurt.** Each added step is another estimate with its own error, and the errors add up: the best five-step water chain scored 15.0 dB on set C, 1.8 dB below the best one-step chain (16.9 dB) and only 0.7 dB above doing nothing (14.3 dB).

The same test on real pictures [REAL]. Real haze (from the 185 pairs: sets of 24, 23 and 46 pairs) and real tank water (TUB; sets of 31, 30 and 59 pairs), with the same pool of steps and the same design.

pictures	set	0	1	2	3	4	5	length chosen on B
real haze	B	12.37	14.50	15.09	15.60	15.22	15.31	3
real haze	C	12.43	13.83	14.72	15.06	14.69	14.34	
real tank water	B	19.01	19.10	19.20	16.81	14.98	13.70	2
real tank water	C	17.94	17.99	18.09	16.11	14.33	13.43	

Table 22: [REAL] Held-out PSNR of the best chain of each length (0 = doing nothing). B chose the length; C was never used.

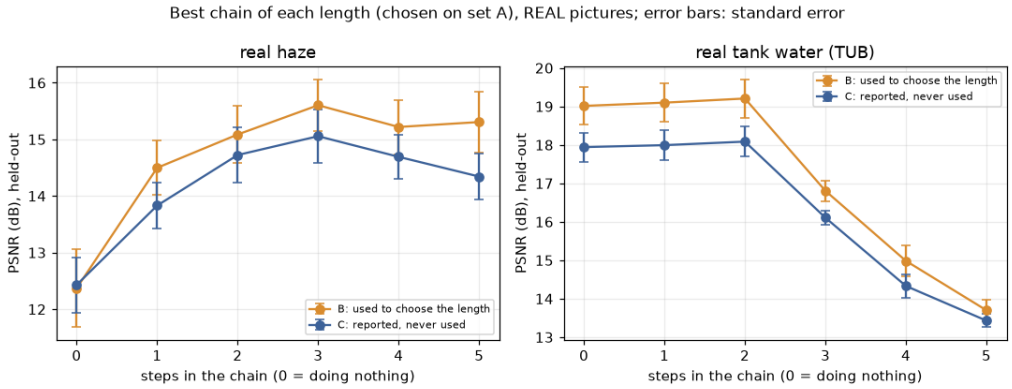


Figure 34: [REAL] Best chain of each length. Error bars: standard error over pictures.

- **Real haze peaks at 3 steps and then stops.** Going from 2 to 3 steps still gains $+0.52 \pm 0.17$ dB (B) and $+0.33 \pm 0.08$ (C); the 4th step gains -0.38 ± 0.39 and -0.36 ± 0.27, the 5th $+0.09 \pm 0.14$ and -0.35 ± 0.08. The best 3-step chain is `dcp_guided > gray_world > clahe` (white balance between the haze removal and the contrast step). That a third step still pays is not a contradiction of the 1-to-2-step theory: real haze is tinted, which our model does not include, and the gray-world step undoes that tint.
- **Real tank water: 2 steps are as far as it goes, and the gain is tiny.** `red_comp > unsharp` gains 0.2 dB over doing nothing (19.2 against 19.0 dB on B). The 3rd step loses 2.4 ± 0.3 dB, the 4th 1.8 ± 0.3, the 5th 1.3 ± 0.3: three or more steps are worse than doing nothing.

The prediction held on real pictures too: chains of 4 and 5 steps do not help. The useful range is up to 3 steps for real haze and 2 for real tank water.

Conclusion: chains of 2 to 3 steps are the useful range, on synthetic and on real pictures; the damage is invertible in one to two steps and the remaining headroom comes from estimating the parameters better, not from adding steps.

9.2 Haze removal by subtraction

Haze adds a smooth glow. Could we simply estimate that glow and subtract it? A *constant* glow does not work (11.4 dB, worse than doing nothing at 12.2). A *local* glow that follows the depth, estimated from how much the detail has faded, reaches **15.0 dB on real haze** (guided DCP: 13.9; the best possible simple frequency filter: 15.4, an upper bound). The lesson: the veil depends on distance, so it must be estimated locally. One caution: the settings of this method came from the same four datasets, so it needs testing on other scenes.

data	method	pictures	PSNR dB	SSIM
real	no restoration	91	12.24	0.488
real	subtract low frequencies (sigma 0.05)	91	12.14	0.604
real	subtract low frequencies (sigma 0.10)	91	12.56	0.619
real	subtract low frequencies (sigma 0.20)	91	12.79	0.618
real	invert model, constant veil	91	11.42	0.481
real	invert model, local veil	91	15.04	0.634
real	homomorphic	91	12.61	0.604
real	frequency Laplacian	91	12.27	0.516
real	oracle radial filter (upper bound)	91	15.43	0.625
real	DCP (guided)	91	13.85	0.52
real	DCP then CLAHE	91	14.66	0.634
twin	no restoration	91	12.37	0.539
twin	subtract low frequencies (sigma 0.05)	91	12.53	0.623
twin	subtract low frequencies (sigma 0.10)	91	12.02	0.619
twin	subtract low frequencies (sigma 0.20)	91	11.07	0.604
twin	invert model, constant veil	91	13.76	0.606
twin	invert model, local veil	91	15.03	0.661
twin	homomorphic	91	10.54	0.591
twin	frequency Laplacian	91	12.4	0.567
twin	oracle radial filter (upper bound)	91	15.44	0.664
twin	DCP (guided)	91	15.94	0.698
twin	DCP then CLAHE	91	15.8	0.742

Table 23: Haze removal by subtraction on the held-out real pairs ("real", **[REAL]**) and on their synthetic twins ("twin", **[SYNTHETIC]**).

9.3 Subtraction combined with the other tools

Subtraction was first tried alone. We then put both subtraction methods into the pool with the DCP family and the contrast tools, and tried every chain of up to three different steps (9 steps, 585 chains), choosing the best on one set of pictures and judging it on another. For real haze we used only the half of each dataset that was *not* used to measure the method's two priors, so the test is not contaminated. (An earlier run that mixed the halves made the subtraction method look 0.7 dB better, so we threw it away.)

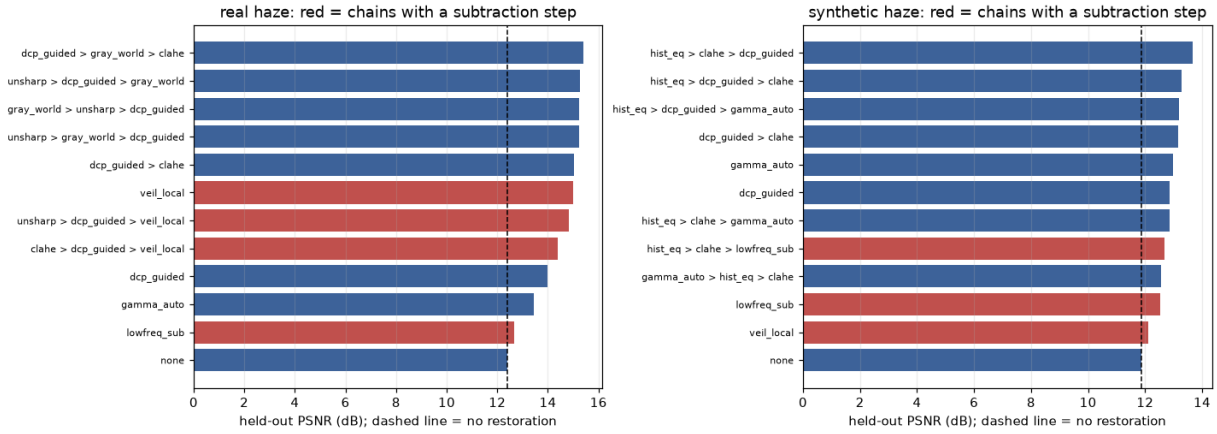


Figure 35: Held-out PSNR of the best chains and some references (dashed line: no restoration). Red bars contain a subtraction step. Left: **[REAL]** haze (23 pairs); right: **[SYNTHETIC]** calibrated haze (18 pictures).

- **On real haze, subtraction alone is as good as anything:** 14.98 dB, tied with DCP then CLAHE (15.02). The best chain overall, `unsharp > gray_world > dcp_guided` (15.24), does not use it, and the best chain that does use it is slightly lower (14.84). So *no combination with subtraction beat the best combination without it.*
- **Where in the chain it goes matters.** Averaged over all chains, a subtraction step placed *last* helps (13.7 dB for three-step chains), placed *first* does not (12.4, the same as chains without it). Subtraction inverts the haze equation, so it works best on a picture that has not been altered; filters applied before it break the equation it assumes.
- **On synthetic haze the order of the winners reverses:** DCP-based chains win (13.66) and subtraction chains lose (12.67). This is the same real-versus-synthetic disagreement as before, now for a method family.
- One chain (`blind_veil_local > tarel > dcp_guided`) produces invalid pictures and was dropped.

9.4 Why DCP fails on water

DCP guesses the glow from the brightest pixels. In water the true glow is a dark blue-green (about 0.34) but bright objects fool DCP into guessing 0.84. An error of 0.50 in A spoils every pixel. Colour methods avoid this because they need no glow estimate.

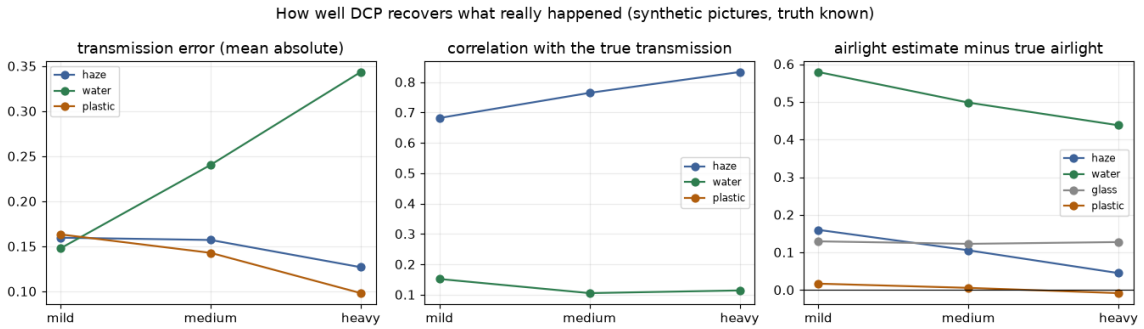


Figure 36: [SYNTHETIC] How well DCP finds the transmission t for haze, water and plastic (correlation with the truth: +0.76 haze, +0.11 water).

10 How good can a repair be? Equations, refinements and limits

This section sets our equations beside the ones in the literature, asks what the real pictures say should be added, writes the inverse of every function we use, and explains why the measured scores sit far below the best possible.

10.1 General equations and ours

Entries marked † are quoted from memory of the papers and should be checked before they are cited.

Medium	Common equation in the literature	Ours
Air	Atmospheric scattering model (Koschmieder; McCartney; Narasimhan and Nayar): $I = Jt + A(1-t)$, $t = e^{-\beta d}$. Variants: a β_c per colour, a $\beta(x)$ that varies over the picture, a glow $A(x)$ for night scenes, a forward-scatter blur.	The same equation, with one scalar β and one scalar glow A.
Water	Jaffe–McGlamery: total light = direct + forward scatter + backscatter, where forward scatter is the direct signal blurred. Simplified form (Chiang and Chen): $I_c = J_c e^{-\beta_c d} + B_c(1-e^{-\beta_c d})$.† Revised (Akkaynak and Treibitz): the direct and the backscatter coefficients differ, $\beta^D \neq \beta^B$.†	The simplified form with a β_c per colour. No forward scatter, and $\beta^D = \beta^B$.
Plastic	ASTM D1003 haze: diffuse light (scattered by more than 2.5°) as a share of the total transmitted light. Imaging models: a point-spread function with a narrow core and a wide halo, plus attenuation and veil.	A Gaussian blur, a tint and a milky veil. No wide halo, no scratches.
Glass	Raindrops (Qian et al.): $I = (1 - M) \odot B + R$, a mask blend.† Dirt (Eigen et al.): an opacity-weighted dirt layer.† Veiling glare: stray light $I = J + g\,(K * J)$ with a wide kernel K; Stiles–Holladay $L = 10E/\theta^2$.†	A film as haze with a depth-free t, smudges as a blur blend, droplets as a blend with a blurred scene. Real drops refract an inverted miniature of the scene, which a blur only imitates.

10.2 What the real pictures say to add

For every real pair we fitted four nested models of how the clear picture became the degraded one, and scored how well each reproduces the real degraded picture (PSNR of the fitted picture; higher means the model explains the real picture better).

pictures	n	one t and A	per colour	+ blur	+ local maps	blur width	t (R/G/B)	glow A (R/G/B)
real haze	185	21.52	22.85	23.04	31.26	0.0132	0.34/0.30/0.26	0.61/0.63/0.69
real raindrops	58	26.08	26.14	26.14	27.0	0.0	0.84/0.85/0.84	0.44/0.46/0.45
real tank water	119	25.05	28.52	29.75	36.7	0.0276	0.30/0.29/0.27	0.57/0.53/0.42

Table 25: [REAL] Fit of nested models to the real pairs. "One t and A": $I = aJ + b$. "Per colour": a separate a_c, b_c per channel. "+ blur": the scene is blurred first (best width per pair). "+ local maps": a separate map in each cell of a 4×4 grid (96 parameters per pair, so part of this gain is flexibility). t and A are the means of the per-colour fit.

- **Haze: the biggest missing piece is that t varies across the picture** (+8.2 dB), which means depth. Colour dependence adds +1.3 dB: blue is attenuated most ($t = 0.34, 0.30, 0.26$ for red, green, blue) and the glow is bluish ($A = 0.61, 0.63, 0.69$; our first model used 0.92 for all colours). Blur adds only +0.2 dB.
- **Milky tank water: the colour is in the glow, not in the attenuation.** The glow is yellow-brown ($A = 0.57, 0.53, 0.42$) while t is nearly the same for all colours (0.30, 0.29, 0.27), which is the opposite of the open-sea model. Per colour +3.5 dB, blur +1.2 dB (width 2.8% of the picture), uneven light over the picture +7.0 dB (the lamp).
- **Raindrops: none of the affine refinements explains them** (26.1 to 27.0 dB). They need a patch model with refraction.

We then built the refined haze (per-colour β and glow, plus the blur, fitted with the values above and calibrated to the real severity as before) and tested, as in Test 7, whether it ranks the methods like real haze does.

pictures	n	Spearman vs real haze	best method	PSNR	guided DCP	none
indoor rooms, original haze (synthetic)	90	+0.57	dcp_guided	15.63	15.63	10.16
indoor rooms, calibrated haze (synthetic)	90	+0.30	hist_spec	13.46	12.42	11.28
indoor rooms, refined haze: per-colour t and glow, blur (synthetic)	90	+0.28	hist_spec	13.14	12.16	11.21
indoor rooms, uniform fog, no depth (synthetic)	90	-0.12	stretch	47.72	18.68	12.88
outdoor, RESIDE-made haze (synthetic)	100	+0.09	stretch	23.83	18.68	16.01

Table 26: [SYNTHETIC] Rank agreement of five synthetic hazes with the 185 real haze pairs (17 methods), and the winners.

- **The refinements did not improve the ranking** (+0.28, against +0.30 for the first calibrated haze and +0.57 for the original haze). Their picture statistics are no closer either (signature distance 0.35, 0.27, 0.42 against 0.34, 0.23, 0.17).
- **A uniform fog is the worst** (−0.12): its damage is one global affine map, so a plain stretch undoes it almost exactly (47.7 dB), which is not what happens on real haze. This confirms that the dependence on depth matters.
- So the missing ingredient for matching the ranking is not colour or blur. We have not found it: the original haze, which looks least like the real fog in its statistics, ranks the methods closest to real haze.

10.3 The inverse of every function we use

Every limit is also limited by the rounding of the saved picture to 8 bits, which each division by t multiplies: $\mathrm{PSNR} \le -10 \log_{10}\left(\overline{1/t^2}/(12 \cdot 255^2)\right)$.

Function	Inverse it computes	Theoretical limit
`stretch`	$J = (I - b)/a$, exact for one global affine map	Fails where t varies across the picture.
`gamma`, `log`, `hist_eq`, `hist_spec`, `eq_hsi`	The inverse of a monotone map applied to the pixel values (equalisation needs the clear picture's histogram)	Ceiling: the best possible map per colour, the *oracle LUT*.
`gray_world`, `white_patch`, `von_kries`	$J_c = I_c/g_c$	Ignores the glow offset and relies on a scene assumption.
`red_comp`	A heuristic, $R' = R + (\bar{G} - \bar{R})(1 - R)G$; the true inverse is $R = R_{\mathrm{obs}} e^{\beta_R d}$, which needs the depth	Restores only the mean of red.

Function	Inverse it computes	Theoretical limit
`dcp`, `dcp_guided`	$J = \dfrac{I - A}{\max(t, t_0)} + A$, with $t = 1 - \omega \operatorname{dark}(I/A)$	Exact if t and A are right. With $t_0 = 0.1$ noise is amplified up to 10× (a cap of about 39 dB).
`udcp`	The same, with the dark channel from green and blue	Needs red already lost.
`tarel`	$J = (I - V)/(1 - V)$	Assumes a white glow.
`blind_veil`	$J = (I - \hat{V})/\hat{t}$	Assumes the local clear brightness is near the prior mean.
`clahe`	A local monotone map; no unique inverse	Noise grows with the clip limit.
`unsharp`	$I + k(I - h*I)$, the first term of the series that inverts a blur	Amplifies high-frequency noise.
`wiener`, `cls`, `richardson_lucy`	$\hat{J} = \dfrac{H^*}{\lvert H\rvert^2 + K}\hat{I}$ for a Gaussian blur, $H = e^{-2\pi^2\sigma^2 f^2}$	Detail above $f^* \approx \sqrt{\ln \mathrm{SNR}/(2\pi^2\sigma^2)}$ is lost.
`flat_field`	$J = I/s$ with s a wide blur	Also removes true slow scene content.
`homomorphic`	Inverts $I = s \cdot r$ in the log domain	Needs the lighting to be purely low-frequency.
`fusion`	A weighted blend, not an inverse	The output stays between its inputs.

For the Wiener limit, rounding noise of 0.0011 and a signal of 0.2 give $f^* \approx 0.51/\sigma$ (an estimate): our plastic blur ($\sigma \approx 1.5$ pixels) is nearly invertible up to 0.34 cycles per pixel, the heaviest ($\sigma \approx 3$) only up to 0.17.

How the limits combine in a chain. (1) A chain of pointwise steps is one monotone map per colour, so its ceiling is the oracle LUT however long it is. (2) Affine layers compose, $J = (I - b_{\mathrm{tot}})/a_{\mathrm{tot}}$ with $a_{\mathrm{tot}} = \prod t_i$ and $b_{\mathrm{tot}} = \sum_i A_i(1 - t_i)\prod_{j>i} t_j$; the last layer is undone first. An affine map commutes with a normalised blur, so in an ideal world their order does not matter; in practice it does, because each step is an estimate and clipping to $[0, 1]$ breaks the commutation.

10.4 Why the measured scores are far below the best possible

Table 20 (Section 9.1) shows chains at 13 to 18 dB against true inverses at 30 to 46 dB. The gap has four parts, which we separate by giving the method more and more of the truth.

- **The per-pixel transmission.** A whole-picture step cannot know that far pixels are hazier. At medium severity the haze chain reaches 13.4 dB and the best global affine map (which uses the clear picture) 14.9, the best global map of any shape (the LUT) 19.7, and the true per-pixel inverse 33.2 dB. The step from the LUT to the true inverse (+13.5 dB for haze, +26 dB for water) is the cost of $t(x)$.
- **Estimating the glow.** The table below gives DCP the truth piece by piece on synthetic haze. With the true glow, DCP rises from 12.9 to 15.0 dB. With the true t but the *estimated* glow, the result is 10.8 dB, worse than DCP itself: an error ΔA in the glow becomes an error $\Delta A/t$ in the picture.
- **Estimating t.** With the true glow, DCP's own t gives 15.0 dB and the true t gives 22.3 dB (with the floor t_0): the dark-channel guess of t costs 7.3 dB.
- **Rounding and the blur.** At heavy haze the true inverse itself is poor (14.3 dB, below the 18.2 dB of the oracle LUT), because t is tiny and the 8-bit rounding is multiplied by $1/t$. For plastic, inverting the blur costs about 5 dB (oracle affine 23.9 and true inverse 29.1 at medium).

severity	none	DCP	DCP, true glow	true t, estimated glow	true t and glow, t floor 0.1	true inverse
mild	14.1	14.2	18.7	10.8	36.0	45.3
medium	11.9	12.9	15.2	11.4	18.5	33.2
heavy	9.5	11.6	11.1	10.3	12.4	14.3
all	11.9	12.9	15.0	10.8	22.3	30.9

Table 28: [SYNTHETIC] DCP given the truth piece by piece (calibrated synthetic haze, 18 held-out pictures; PSNR in dB).

Are any results almost the best possible? Yes, in two situations. At *heavy haze* the chain reaches 12.4 dB against a true inverse of only 14.3 dB, and in 5 of the 18 pictures the best chain is within 3 dB of the true inverse. For *raindrops* the best chains are within 1 dB of the oracle LUT in 10 of 18 pictures, but that ceiling is

only 0.9 dB above doing nothing, so there is nothing to gain. For haze and water at mild and medium severity no chain comes within 1 dB of the LUT.

10.5 Why the gaps are so large, and which pictures are beyond repair

Six things separate a measured score from the theoretical one, roughly from the most to the least important.

1. **The theory is told the truth; the method is not.** The true inverse uses the true $t(x)$ for every pixel, the true glow and the true blur. A method must guess them from the damaged picture alone. For $t(x)$ this is an ill-posed problem (a bright grey patch may be a hazy far wall or a near white wall), so even a good guess is wrong in places, and a wrong t or A is magnified by $1/t$.
2. **The priors fail on some pictures.** DCP assumes a dark pixel in every patch (the clear pictures have a mean dark channel of 0.12 to 0.21, not 0), a stretch assumes the clear picture spans the full range, white balance assumes a grey scene.
3. **The information may be gone.** Rounding to 8 bits (and sensor noise in real pictures) is multiplied by $1/t$; where t is small the signal is buried. This is the only case where no method, however clever, can succeed.
4. **The model is wrong for real pictures.** Real haze has a coloured glow and varies over the picture; real raindrops replace patches (Section 10).
5. **PSNR punishes brightness and colour shifts** that look acceptable, so an equally good-looking restoration scores lower.
6. **The functions and their settings may be incomplete**: the plastic result of Section 10.7 (+8 dB from a per-colour stretch and the true blur width).

Beyond repair. We call a synthetic picture *beyond repair* when more than half of its pixels have $t < 0.0055$: there the rounding error of 0.0011, multiplied by $1/t$, exceeds a typical scene contrast of 0.2, so the signal-to-noise ratio is below one and not even the exact inverse can bring the pixel back. For real pictures t is unknown, so we use what is left of the scene's detail (the sharpness ratio: the detail of the degraded picture relative to the clear one) and ask whether any method we tried restores the structure (SSIM) and not just the brightness.

medium	severity	pixels unrecoverable	beyond repair	near the theory	in between	estimation-limited (gap 10 dB or more)
haze	mild	0%	0	0	0	18
haze	medium	0%	0	0	0	18
haze	heavy	46%	6	11	1	0
water	mild	0%	0	0	0	18
water	medium	0%	0	0	0	18
water	heavy	0%	0	0	0	18

Table 29: [SYNTHETIC] Classes of the 18 held-out pictures per medium and severity. "Near the theory" means the best chain is within 3 dB of the true inverse, or above it (heavy haze, where the true inverse itself fails).

- **Haze, mild and medium, and water at every severity, are estimation-limited:** all 18 pictures in each row have a true inverse 10 dB or more above the best chain. Nothing is lost; the methods simply cannot see $t(x)$. (Our synthetic water never loses pixels, since its largest attenuation, $e^{-4.8} = 0.008$, stays above the limit.)
- **Heavy synthetic haze contains pictures beyond repair:** on average 46% of the pixels are unrecoverable; 6 of the 18 pictures are beyond repair, and for 11 more the best chain already matches or beats the (failing) inverse.

pictures	detail left (sharpness ratio)	n	mean best gain (dB)	share with no gain above 0.5 dB
real haze	0.0 to 0.1	38	4.73	5%
real haze	0.1 to 0.3	72	4.14	1%
real haze	0.3 to above	75	4.25	5%
real tank water	0.0 to 0.1	39	0.45	72%
real tank water	0.1 to 0.3	59	0.68	54%
real tank water	0.3 to above	22	0.26	86%

Table 30: [REAL] What the best of the methods tried can do, by how much detail is left (sharpness ratio). Real haze: 185 pairs; real tank water: 120 pairs.

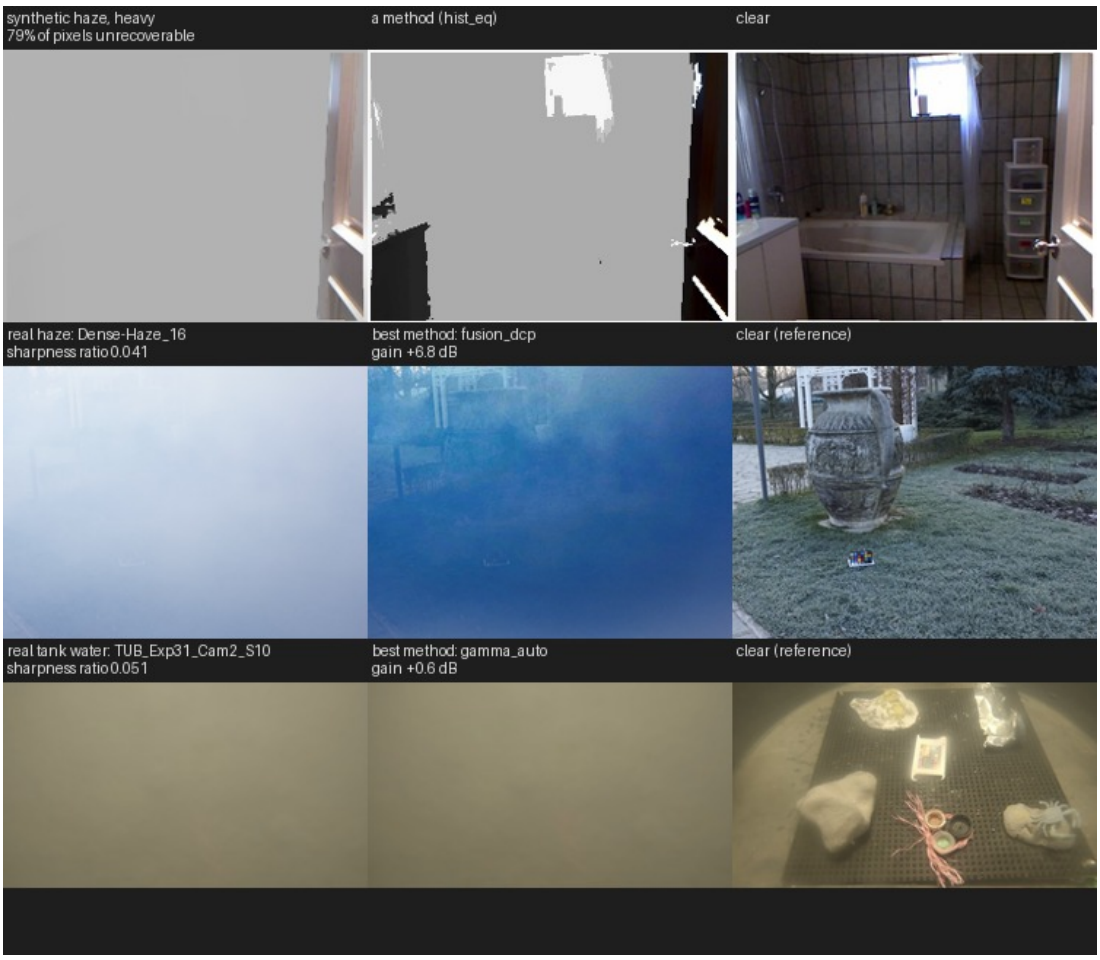


Figure 37: Pictures beyond repair. Row 1: **[SYNTHETIC]** heavy haze, 79% of the pixels unrecoverable (histogram equalisation shows flat grey where the information is gone). Row 2: **[REAL]** Dense-Haze, the lowest detail left (sharpness ratio 0.04); the best method (fusion after DCP) gains 6.8 dB, but the result is a blue wash in which the vase is not back. Row 3: **[REAL]** tank water at the highest turbidity: a white-out; the best method gains 0.6 dB and recovers nothing.

- **Real Dense-Haze: 38 of its 55 pairs are close to beyond repair.** The sharpness ratio is below 0.1 for all 38. Their PSNR still improves by 4.7 dB, but SSIM only rises from 0.30 to 0.46, against 0.65 to 0.80 for the lighter hazes: brightness and colour are restored, the structure only in part. The other three datasets (O-HAZE, I-HAZE, NH-Haze) have no pair below 0.1.
- **Real tank water: no picture is repaired.** The best of 17 methods gains about 0.5 dB on average (0.2 to 0.7 dB across turbidity bands), SSIM stays at 0.665 to 0.670 for the densest 39 pairs (the white-outs), and 54 to 86% of the pairs gain less than 0.5 dB in each detail band. At low turbidity there is little to repair (the unrepaired picture already scores 21 dB); at high turbidity the detail is gone.
- **Raindrops** are a different case: the covered patches are gone, but the rest of the picture is fine, so the picture is not beyond repair but beyond *global* repair; a method that detects and fills drops (inpainting) is outside this study.

10.6 The best chain against the theoretical restoration, in pictures

For every synthetic held-out picture we took the best of all chains tried (the earlier searches and the chains derived from the inverse in the next subsection; chosen with the answer known, so slightly optimistic) and compared it with the theoretical restoration: the true inverse, or for raindrops (which have no inverse) the best possible map of pixel values, the oracle LUT. The three rows of each figure are the case with the smallest gap, a typical (median) gap and the largest gap. Columns: degraded, best chain, oracle LUT, theoretical restoration, clear.

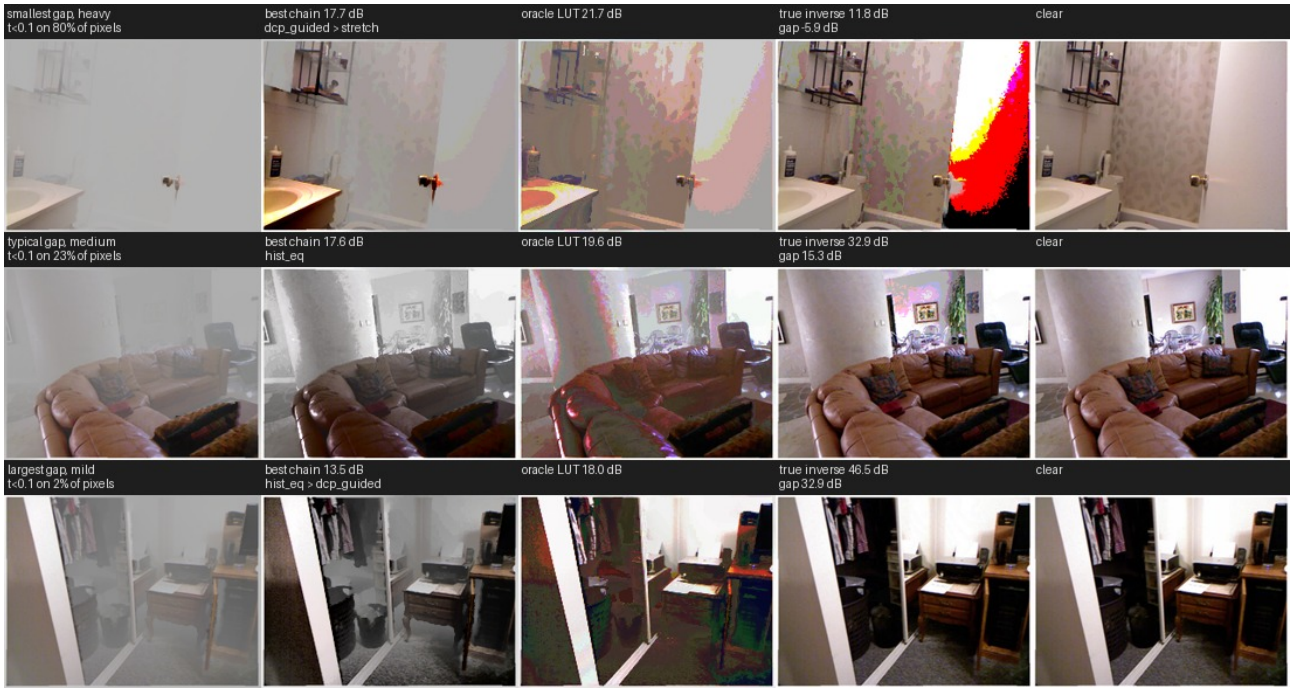


Figure 38: [SYNTHETIC] Haze. Top: heavy haze, where the best chain beats the true inverse. Middle: medium haze. Bottom: mild haze, the largest gap.

- **Smallest gap, heavy haze: the chain beats the theory (-5.9 dB).** Here $t < 0.1$ on 80% of the pixels, so the 8-bit rounding of the picture is multiplied by $1/t$ (up to 10^8 in $1/t^2$) and the true inverse breaks down (saturated white and red patches). A chain that raises contrast without dividing by t is better.
- **Typical gap, medium haze (15 dB).** The true inverse is almost exact (32.9 dB); the chain (histogram equalisation, 17.6 dB) brightens the whole picture but cannot know that the far wall is hazier than the near sofa, so a veil remains where t is small.
- **Largest gap, mild haze (33 dB).** Only 2% of the pixels have $t < 0.1$, so the true inverse is nearly perfect (46.5 dB), but any chain that cannot see $t(x)$ leaves the depth gradient (the grey wash at the back of the room) and cannot restore the colours.



Figure 39: [SYNTHETIC] Open-sea style water, same layout.

Water behaves like haze: the gap is small only when rounding limits the inverse (heavy: $t < 0.1$ on 78% of the pixels, true inverse 29.7 dB, gap 14 dB), and grows to 30 to 38 dB at medium and mild severity where the inverse is almost exact (49 to 55 dB) but the chain's colour correction is global (17.5 to 18.5 dB).

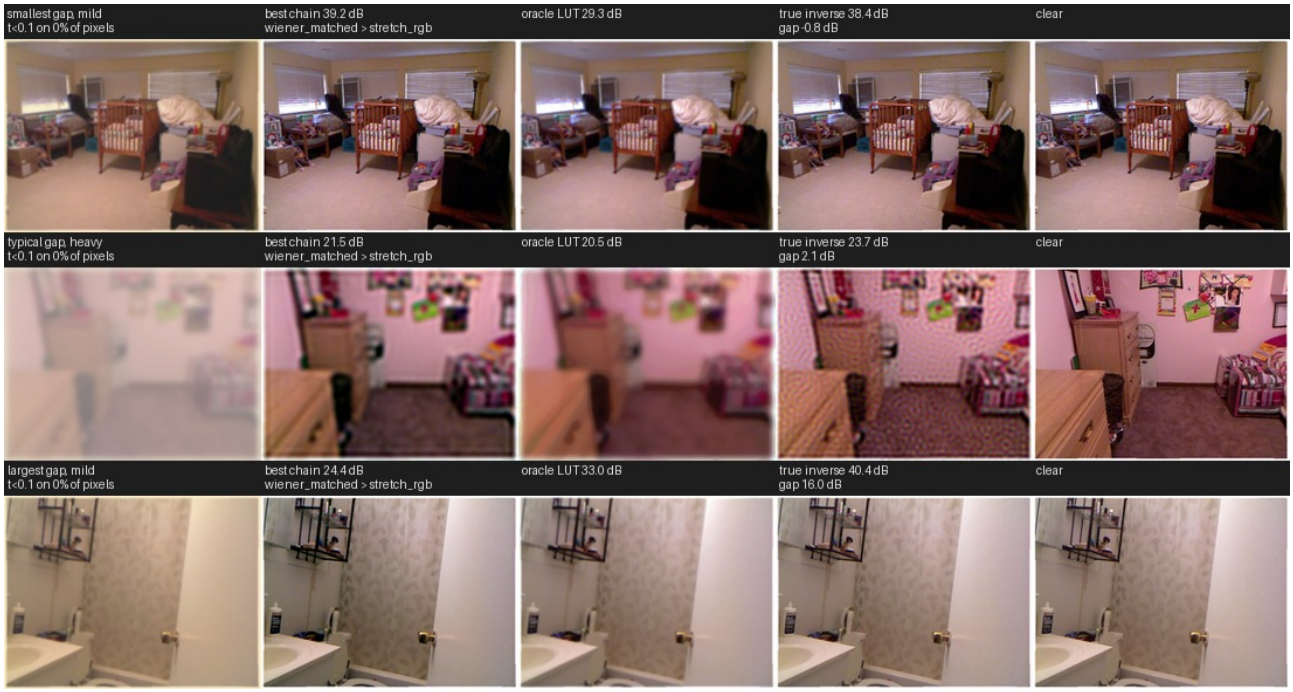


Figure 40: [SYNTHETIC] Plastic. The best chain is the derived `wiener_matched > stretch_rgb` in all three cases.

- **Smallest gap, mild plastic (-0.8 dB):** the chain is essentially the inverse (undo the blur, then one per-colour range), and it matches the true inverse (39.2 against 38.4 dB).
- **Typical gap, heavy plastic (2.1 dB):** a 3-pixel blur; the Wiener filter cannot recover detail above about 0.17 cycles per pixel (Section 10), so the result stays softer than the true inverse.
- **Largest gap, mild plastic (16 dB):** the blur is easy, but the per-colour stretch *assumes the clear picture spans the full range from black to white*. This bathroom is bright with few deep shadows, so the likely cause (not tested) is that the stretch maps the wrong range and leaves a brightness and colour offset; the true inverse knows the true gain and offset.

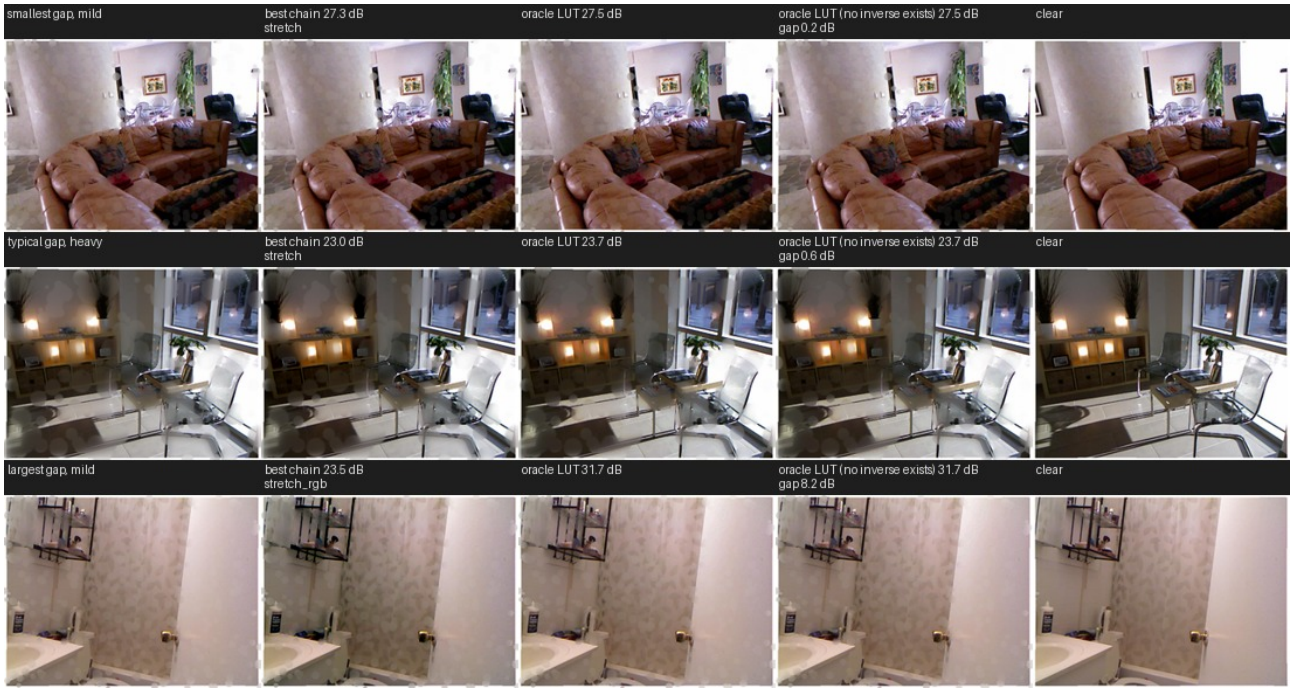


Figure 41: [SYNTHETIC] Raindrops. The theoretical column is the oracle LUT (no inverse exists).

For raindrops the typical and smallest gaps are only 0.2 to 0.6 dB: the best chain (a plain stretch) is already close to the best possible whole-picture map, and that map is only about 1 dB above doing nothing. The largest gap (8 dB) is a picture where the oracle map, which sees the answer, does much better than any rule that has to guess; we did not look into why.

10.7 Chains derived from the inverse equations, and whether we had tried them

The inverse of each medium is a short list of operations (undo the gain and offset per colour; undo the depth-dependent transmission; undo the blur), and each operation has a function that approximates it. We built the chains in the order the inverse asks for and tested them next to the best chains of the earlier searches. One function was missing: the stretch in the earlier searches used a single range for all colours, so it could not undo a different gain and offset per colour. We added `stretch_rgb` (a separate range per colour), and a Wiener filter whose blur width equals the true blur of the severity (`wiener_matched`).

pictures	chain	origin	tried before?	PSNR	vs best earlier
synthetic haze	none	–	–	11.61	-3.22 ±
synthetic haze	dcp_guided > stretch	derived from the inverse	no	12.94	-1.89 ±
synthetic haze	gray_world > dcp_guided	derived from the inverse	yes	12.95	-1.89 ±
synthetic haze	gray_world > dcp_guided > stretch	derived from the inverse	no	12.83	-2.00 ±
synthetic haze	dcp_guided > gray_world	derived from the inverse	yes	12.98	-1.85 ±
synthetic haze	dcp_guided > gray_world > stretch_rgb	derived from the inverse	no	12.89	-1.94 ±
synthetic haze	tarel > stretch	derived from the inverse	no	10.92	-3.91 ±
synthetic haze	blind_veil_local > stretch	derived from the inverse	no	12.16	-2.68 ±
synthetic haze	stretch_rgb > dcp_guided	derived from the inverse	no	12.29	-2.55 ±
synthetic haze	dcp_guided > gamma_auto	derived from the inverse	yes	13.54	-1.29 ±
synthetic haze	hist_eq	earlier search	–	13.07	-1.76 ±
synthetic haze	hist_eq > dcp_guided	earlier search	–	14.54	-0.30 ±
synthetic haze	hist_eq > clahe > dcp_guided	earlier search	–	14.83	+0.00
synthetic haze	ceilings: oracle LUT / true inverse			19.4 / 30.0	
synthetic water	none	–	–	14.64	-2.76 ±
synthetic water	stretch_rgb	derived from the inverse	no	14.82	-2.58 ±
synthetic water	gray_world > stretch_rgb	derived from the inverse	no	15.32	-2.08 ±
synthetic water	red_comp > stretch_rgb	derived from the inverse	no	15.85	-1.55 ±
synthetic water	stretch_rgb > von_kries	derived from the inverse	no	16.81	-0.59 ±
synthetic water	udcp > stretch_rgb	derived from the inverse	no	11.74	-5.66
synthetic water	red_comp > von_kries > stretch_rgb	derived from the inverse	no	16.73	-0.67 ±
synthetic water	von_kries > stretch_rgb	derived from the inverse	no	16.38	-1.01 ±

pictures	chain	origin	tried before?	PSNR	vs best earlier
synthetic water	von_kries	earlier search	–	17.17	-0.23 ±
synthetic water	red_comp > von_kries	earlier search	–	17.35	-0.05 ±
synthetic water	red_comp > von_kries > unsharp	earlier search	–	17.40	+0.00
synthetic water	ceilings: oracle LUT / true inverse			21.2 / 45.5	
synthetic plastic	none	–	–	15.81	-2.33 ±
synthetic plastic	stretch > wiener	derived from the inverse	no	20.26	+2.12 ±
synthetic plastic	wiener > stretch	derived from the inverse	no	20.41	+2.28 ±
synthetic plastic	stretch_rgb > wiener	derived from the inverse	no	21.99	+3.85 ±
synthetic plastic	wiener > stretch_rgb	derived from the inverse	no	21.73	+3.60 ±
synthetic plastic	stretch_rgb > richardson_lucy	derived from the inverse	no	22.17	+4.04 ±
synthetic plastic	stretch_rgb > wiener_matched	derived from the inverse	no	26.01	+7.87 ±
synthetic plastic	wiener_matched > stretch_rgb	derived from the inverse	no	26.57	+8.43 ±
synthetic plastic	clahe	earlier search	–	18.14	+0.00
synthetic plastic	wiener > dcp_guided	earlier search	–	17.68	-0.45 ±
synthetic plastic	wiener > unsharp > dcp_guided	earlier search	–	17.76	-0.38 ±
synthetic plastic	ceilings: oracle LUT / true inverse			24.2 / –	
synthetic glass	none	–	–	25.36	+0.50 ±
synthetic glass	flat_field > stretch	derived from the inverse	yes	19.58	-5.28 ±
synthetic glass	stretch > flat_field	derived from the inverse	yes	19.59	-5.27 ±
synthetic glass	flat_field > stretch_rgb	derived from the inverse	no	19.55	-5.31 ±
synthetic glass	stretch_rgb	derived from the inverse	no	24.67	-0.19 ±
synthetic glass	stretch	earlier search	–	24.86	+0.00
synthetic glass	stretch > unsharp	earlier search	–	22.64	-2.22 ±
synthetic glass	flat_field > unsharp > stretch	earlier search	–	19.28	-5.58 ±
synthetic glass	ceilings: oracle LUT / true inverse			26.2 / –	
real haze	none	–	–	12.41	-2.83 ±
real haze	dcp_guided > stretch	derived from the inverse	no	12.70	-2.54 ±
real haze	gray_world > dcp_guided	derived from the inverse	yes	14.29	-0.95 ±
real haze	gray_world > dcp_guided > stretch	derived from the inverse	no	13.08	-2.15 ±
real haze	dcp_guided > gray_world	derived from the inverse	yes	14.39	-0.85 ±
real haze	dcp_guided > gray_world > stretch_rgb	derived from the inverse	no	13.30	-1.94 ±
real haze	tarel > stretch	derived from the inverse	no	9.81	-5.43 ±
real haze	blind_veil_local > stretch	derived from the inverse	no	13.82	-1.42 ±
real haze	stretch_rgb > dcp_guided	derived from the inverse	no	13.47	-1.76 ±
real haze	dcp_guided > gamma_auto	derived from the inverse	yes	13.24	-2.00 ±
real haze	dcp_guided > gray_world > clahe	earlier search	–	15.24	+0.00
real haze	dcp_guided > clahe	earlier search	–	14.85	-0.39 ±
real water	none	–	–	18.30	-0.16 ±
real water	stretch_rgb	derived from the inverse	no	12.86	-5.60 ±
real water	gray_world > stretch_rgb	derived from the inverse	no	12.86	-5.60 ±
real water	red_comp > stretch_rgb	derived from the inverse	no	12.85	-5.61 ±
real water	stretch_rgb > von_kries	derived from the inverse	no	12.61	-5.86 ±
real water	udcp > stretch_rgb	derived from the inverse	no	11.79	-6.67 ±
real water	red_comp > von_kries > stretch_rgb	derived from the inverse	no	12.85	-5.61 ±
real water	von_kries > stretch_rgb	derived from the inverse	no	12.86	-5.60 ±
real water	red_comp > unsharp	earlier search	–	18.46	+0.00
real water	unsharp	earlier search	–	18.36	-0.10 ±

- **Plastic: the derived chains win by a wide margin.** `wiener_matched > stretch_rgb` scores 26.6 dB against 18.1 for the best earlier chain (+8.4 dB), and even with the default blur width `stretch_rgb > richardson_lucy` reaches 22.2 dB (+4.0). The earlier search missed them for two reasons: its pool had no per-colour stretch, and its Wiener filter always assumed one blur width. This chain also lies above the oracle LUT (24.2 dB), as it must: the LUT bounds only chains of pointwise steps, and a deblur is not pointwise.

- **Haze: the derived chains do not beat the earlier ones.** `dcp_guided > gamma_auto` (13.5 dB) and the others stay 1.3 to 3.9 dB below the best earlier chain (14.8 dB), and adding a stretch after DCP lowers the score on real haze (13.1 against 14.4 without it). On real haze the best derived chain, `dcp_guided > gray_world` (14.4), is 0.85 dB below the earlier `dcp_guided > gray_world > clahe`.

- **Water: close, but not better.** `stretch_rgb > von_kries` reaches 16.8 dB on synthetic water, 0.6 dB below the best earlier chain; the per-colour stretch alone (14.8) is worse than white balance (17.2) because the damage varies with depth, which a global range cannot follow. On the real tank water the per-colour

stretch is harmful (12.9 dB against 18.3 for doing nothing): the near-uniform milky picture has almost no range for the percentiles to estimate, so the stretch amplifies noise.

- **Glass:** chains that assume a slow film (`flat_field`) lower the score by 5 dB, as raindrops are not a film.
- **Had we tried them?** Only some. Chains made of the earlier pool's functions at their default settings had been searched (marked "yes"). Chains using `stretch`, `stretch_rgb`, the matched Wiener filter or the subtraction methods inside a medium's own pool had not (marked "no").

So should the functions, their settings and their order change? Yes, where the theory says what the inverse needs: for plastic the settings (the blur width) and a per-colour stretch matter by 4 to 8 dB, more than any reordering. For haze and water the settings are near their optimum (previous subsection) and the derived chains do not beat the searched ones. The order of deblur and stretch commutes in theory; in practice deblurring first is 0.6 dB better (26.6 against 26.0), because the stretch percentiles are estimated on a blurred picture.

10.8 Should the settings and the order of the steps change?

We tuned the settings inside the best real-picture chains on one set of pictures and judged them on two others (the same disjoint sets as in Section 9.1).

pictures	set	default settings	tuned settings	gain (dB)
real haze	A	14.8	14.86	$+0.07 \pm 0.19$
real haze	B	15.6	15.66	$+0.06 \pm 0.25$
real haze	C	15.06	15.26	$+0.20 \pm 0.15$
real tank water	A	19.43	19.97	$+0.54 \pm 0.12$
real tank water	B	19.21	19.68	$+0.48 \pm 0.09$
real tank water	C	18.08	18.41	$+0.33 \pm 0.05$

Table 32: [REAL] Default against tuned settings of the best chains (A chose the settings; B and C are held out). Gain: mean ± standard error, dB.

- **Real haze, `dcp_guided > gray_world > clahe`:** tuning ω, t_0, the patch size, the CLAHE clip and tile count gains $+0.06 \pm 0.25$ dB (B) and $+0.20 \pm 0.15$ (C), within the noise. The defaults are already near the best: the score surface is flat around them.
- **Real tank water, `red_comp > unsharp`:** stronger red compensation (2.0) and a wider unsharp mask ($\sigma = 0.03$) gain $+0.48 \pm 0.09$ (B) and $+0.33 \pm 0.05$ (C), small but consistent. Both best values sit at the edge of the grid, so the best setting may be stronger still.
- **The order matters more than the settings.** The order of the same steps changes the score by up to 3.1 dB (Stage 4d), a subtraction step placed last helps and placed first does not, while tuning the settings gains 0.1 to 0.5 dB. The theory agrees: undo the last layer first, and give each step an unmodified picture to estimate from.

11 Conclusions and limits

1. The one equation $I = tJ + A(1 - t)$ explains all four media; the frequency view confirms it.
2. **DCP is the right tool only for haze,** and even there it is weak on real photos without a following contrast step.
3. **Something beats DCP:** on real haze, DCP then CLAHE (+0.9 dB over DCP alone on all 185 pairs) and local-glow subtraction (+1.2 dB over DCP on its held-out pairs); on synthetic water, colour correction (+5.0 dB over DCP). On real turbid tank water nothing beat doing nothing and DCP was far worse (11.5 against 18.7 dB); on real ocean photos colour correction looks best by eye. Synthetic water made milky (matching the tank) reproduces that verdict; a dark ink cloud, in contrast, is helped by brightening methods.
4. **Order matters,** by up to 3.1 dB, and an extra step can help or hurt.
5. **Made-up pictures help but need checking.** Our first presets gave misleading winners; calibrating to real pictures fixed glass fully and haze partly.

6. **Raindrops cannot be repaired by global methods,** and running methods per tile does not fix that. Nothing beat doing nothing.

7. **For haze, local is better than global:** running histogram equalisation or DCP tile by tile gained 1 to 2 dB on real haze, best at roughly 4×4 to 6×6 tiles; depth is the information that matters. The best tile size differs by medium (water: colour methods like many tiles; glass: none).

8. **Some pictures are beyond repair.** Where more than half of a synthetic picture has $t < 0.0055$ (6 of 18 heavy-haze pictures), the information is gone; on real pictures the densest Dense-Haze (38 pairs) is repaired in brightness but only partly in structure, and the real tank water is not repaired at any turbidity.

9. **The best possible is far above what any chain reaches, and the gap is estimation, not steps.** The true inverse is 13 to 26 dB above the chains at medium severity; its parts are the per-pixel transmission (depth), the estimate of the glow (an error ΔA becomes $\Delta A/t$) and of t, rounding at heavy haze, and the blur for plastic. Real pictures also ask for per-colour glow and local variation, but refining the synthetic haze that way did not make it rank the methods more like real haze. Tuning the settings inside the best chains gains 0.1 to 0.5 dB; the order of the steps matters more.

Limits. Plastic is unchecked against real pictures, and real water is only partly checked (a tank with milky turbidity and a nearly clear reference; ocean photos without a reference; ink has no real data at all). Own captures are still to come. The haze ranking mismatch across scenes is unexplained beyond the scene argument. The subtraction method's settings were fitted on the test data's source datasets.

Reproducibility. Everything is rerunnable stage by stage (`code/experiments/run_stages.py`); a full run takes about 9 minutes on 16 CPU cores. All numbers in this report are read from `results/tables/`. Every finding and failed attempt is in `log.md`.

A Does the frequency theory hold?

case	curve	rms_error	mean_relative_error	measured_mid	predicted_mid
haze_flat	luminance	0.0	0.0	0.301	0.301
haze_depth	luminance	0.103	0.2135	0.561	0.47
haze_depth	luminance_refined	0.0292	0.037	0.561	0.553
plastic	luminance	0.0007	0.0588	0.05	0.05
water_flat	channel 0	0.0	0.0	0.259	0.259
water_flat	channel 1	0.0	0.0	0.592	0.592
water_flat	channel 2	0.0	0.0	0.799	0.799
glass_drops	luminance	0.176	0.234	0.913	0.727
glass_drops	luminance_refined	0.0974	0.1163	0.913	0.812
stack_water_plastic	channel 0	0.0001	0.0575	0.014	0.014
stack_water_plastic	channel 1	0.0003	0.0582	0.03	0.03
stack_water_plastic	channel 2	0.0004	0.0572	0.034	0.034
real raindrops (formula fitted to the data)	luminance	0.0081		0.899	fitted weight 0.18, blur sigma 0.8 px
real haze: spectral ratio against affine-fit t, 185 pairs	correlation			0.789	

Table 33: Predicted against measured detail retention (mid-band ratio). Error 0.0 means exact. "haze_depth" is haze that changes with distance, where a plain formula was off (0.103) and a detail-weighted one fixed it (0.029). The raindrop formula improved from 0.176 to 0.097.

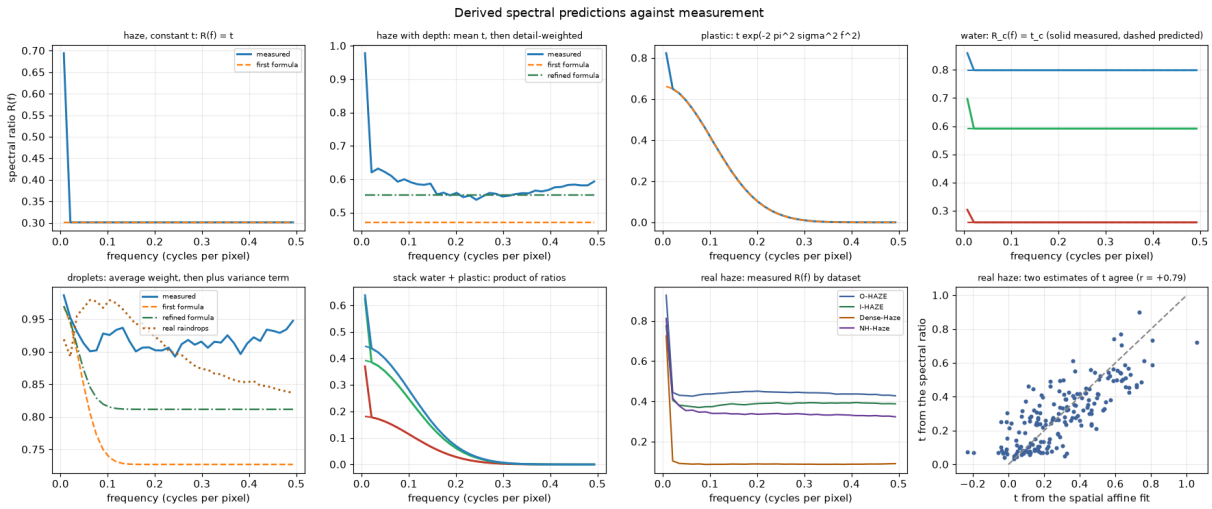


Figure 42: [SYNTHETIC] Measured against predicted detail retention per medium (frequency on the horizontal axis).

B Parameter and severity sweeps [SYNTHETIC]

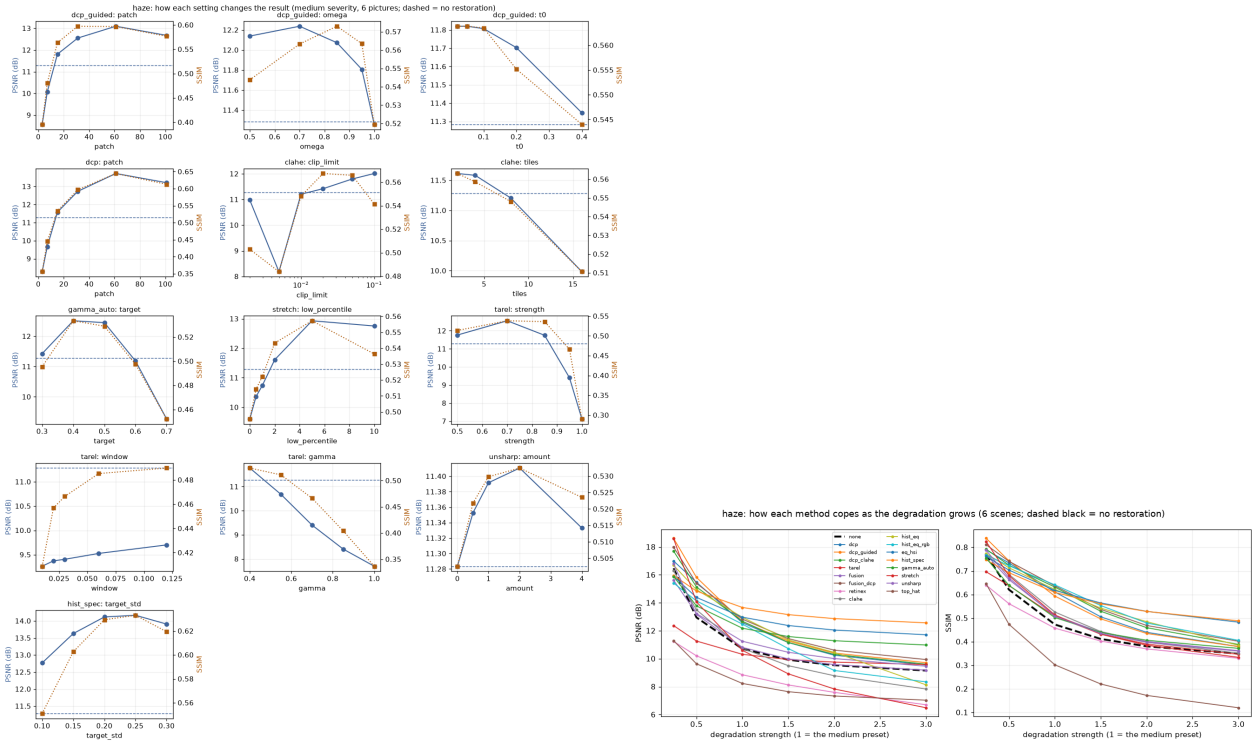


Figure 43: [SYNTHETIC] Left: how much a method's own settings matter (haze). Right: score against fog strength (calibrated haze).